Bitcoin's Required Return

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Abstract

What return should Bitcoin earn? In average annual (arithmetic) terms, its history says 32%, the CAPM 11–16%, its holders say 14–22%. We build what holders should require, adding to the CAPM's market risk two measurable risks it omits: concentration (volatility at holders' large positions) and mortality, measured on 520 cryptoasset deaths. Tested monthly since 2012, it is not rejected: early returns ran above it, and since 2018 average annual returns match it within 10 percentage points. Today it is 18% (90% range 15–22%), or about 9% a year compounded for a buy-and-hold investor, and should keep falling as holders diversify.

Keywords: Bitcoin; cryptoassets; required return; expected return; survival analysis; limited participation; cost of capital; portfolio construction.

1. INTRODUCTION

Suppose you must write down one number: the return Bitcoin should be expected to earn, per year, over the next five years. Three candidate answers are on the table, and they disagree by a factor of three.

The first comes from Bitcoin's own history. A simple rule, the exponent of Bitcoin's power-law price trend, about 5.6 (Burger 2019; Papadogiannis 2026c), divided by Bitcoin's age in years, came closest, among the candidates tested, to an unbiased forecast of the returns that followed: over rolling three- to five-year periods since 2012 its forecast errors averaged nearest to zero, with the exponent re-estimated each month from the data then available (Papadogiannis 2026c). Applied today it says about 32% a year.[1] The second comes from standard asset pricing. The CAPM asks what Bitcoin's risk would cost a diversified investor: the ten-year Treasury yield of 5%, plus Bitcoin's beta to the stock market times an equity premium of 4% (6% as a high case). Bitcoin's beta to the S&P 500, estimated on month-end total returns, is 1.75 over the last four years, 1.5 since 2016, and 1.8 since 2012 (Appendix A), which gives 11–16% across those

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[1] A note on units: the three figures in this paragraph are arithmetic average annual returns. The rule of thumb, exponent over age (β/A), gives 32% at Bitcoin's age in May 2026, the end of our sample ($A = 17.4$ years). Its exact five-year form, $(1+5/A)^{\beta/5}-1$, also gives 32% from September 2026 ($A = 17.7$), the start of the forecast window used throughout, and 33% from mid-2026, the date of the last row of our tables. The compound (geometric) counterpart is lower by roughly half the variance, $\sigma^2/2$, and so depends on the volatility assumed: a drag of about 4.5 points at 30% volatility and 10 points at 45%, which is why the compound figure is a range, 22–28%, and the arithmetic figure a single number.

windows and the two premia. The third comes from asking people. In a large US household survey, those who hold crypto said they expected 22% over the next year in 2021 and 13.8% in 2025 (Candia, Coibion, Gorodnichenko, and Weber 2026); 14–22% is the spread between the two dates.

These are not three estimates of one quantity; the disagreement begins with using one word for three things. The first is an *expected return*: a property of Bitcoin's price process, checkable only against what it later earns. The second is a *normative required return*: what an investor holding the CAPM's diversified portfolio ought to demand for Bitcoin's risk, whether or not any such investor holds it. The third is a *stated required return*: what the people who hold Bitcoin say they expect, an empirical fact about them. What is missing is a fourth, the one this paper builds: the *normative required return of the actual holder*, what the people who hold Bitcoin ought to demand given the risks their own portfolios carry. It is the one the price must satisfy in equilibrium, because they are the ones who must be willing to keep holding. That leaves a puzzle: is the gap between what Bitcoin delivered and what a model says it should deliver payment for risks that did not materialize, or a sign that the market underpriced the asset? Bitcoin's past returns cannot settle this: they are the single history of a survivor. We study Bitcoin because it did not fail, so the risks that could have ended it left no mark on its record.

This paper answers by measuring those risks. It builds a required return of the CAPM's kind, a price for risk derived from a model, but for the holders Bitcoin actually has and for more of the risks they bear. Bitcoin pays no dividend or coupon, so it is worth holding only if its price is expected to rise by enough to pay its holders for the risks they take. We price three.

The first is *market risk*, the risk Bitcoin shares with the stock market, and the one risk the CAPM prices (Sharpe 1964; Lintner 1965). Bitcoin need not belong to a stock index for this to apply: the index stands in for the rest of an investor's wealth, and what is priced is how Bitcoin moves with it. Its beta to the S&P 500 is about 1.75, worth about 7 points a year (throughout, returns are nominal US-dollar returns and "points" are percentage points of arithmetic average annual return), and behind it is a correlation that rose from about 0.2 before 2020 to about 0.5 as institutions entered (Copestake, Furceri, and Terracciano 2026; Appendix A). The second is *concentration risk*. The CAPM assumes every investor is fully diversified, so that an asset's own volatility costs nothing. Bitcoin's holders are not diversified: among US households that own any crypto, crypto makes up 19% of financial assets on average, and one in five keeps more than half its financial assets in it (Candia, Coibion, Gorodnichenko, and Weber 2026). For such a holder Bitcoin's volatility is a cost, and a larger one the larger the share of her wealth she keeps in it, which is how Merton (1987) prices assets held by few. The third is *mortality risk*, on which the CAPM is silent: the chance that the asset dies outright, through a ban, a protocol failure, or a collapse of the kind that ended Terra and FTX. Of the 1,815 cryptoassets in our census, 520 have died (Section 3), and the price of that risk is the standard compensation for a jump to ruin (Merton 1976).

Add the three to the risk-free rate and you have the return the holder requires. Take away the last two and what is left is the CAPM. These are the risks we judge important and can measure; others may exist (Section 9). The first two are measurable from Bitcoin's own prices and from surveys of who holds it. The third cannot be measured from Bitcoin, which has not died; it can only be measured from the deaths of other cryptoassets, as a life insurer prices one person from mortality tables of many.

Mortality risk turns out to be small for an asset like Bitcoin. We estimate the asset class's mortality curves from the census as a demographer would. (Earlier descriptive studies of "coin death" exist, notably

Feder, Gandal, Hamrick, and Moore (2018) and Grobys and Sapkota (2020); new here is a curated, pricing-oriented census: death types, dates, recovery fractions, size- and elite-conditional hazards, and the Lindy test.) The popular "Lindy" intuition, that the longer an asset has survived the safer it is, is rejected: the death rate of cryptoassets *rises* with age. And the hazard that matters for Bitcoin, the death rate of assets that reached the top twenty, is 3.5% a year by any cause and 0.30% a year for sudden, value-destroying collapses (Terra's LUNA and UST and FTX's FTT, all in 2022). That is worth 0.3–3 points of required return. The age rule's 32% is 27 points above the 5% risk-free rate, and the risk of death can account for only a small part of that excess.

Concentration risk was the large term, and it has shrunk. Before the ETF era crypto markets were segmented from mainstream capital (Makarov and Schoar 2020), and the traders who set Bitcoin's price were plausibly far more concentrated than the survey average. Option prices put a floor under what a fully invested holder requires (Martin 2017): 66–84% a year above the risk-free rate in 2017–22. At the 120% volatility of the early years the same holder required more than 100%.[2] Delivered returns were of that order. As volatility fell and the holder base broadened the required return came down, term by measurable term, to about 18% today (90% range 15–22%). Equation (1) in Section 2 adds the three terms to the risk-free rate; we call it the required-return equation. It is Merton's (1987) limited-participation pricing with every input measured, not a factor regression or a consumption-based model, which cannot see the investor who clears this market.

Tested against what Bitcoin delivered, the required return is not rejected. We evaluate it at every month-end since 2012 and compare it with Bitcoin's return over the following month, 172 non-overlapping observations, in matching units: average returns, not compound returns. At each date the required return is a band rather than a point, mainly because the data do not identify who was setting Bitcoin's price: the lower edge assumes survey-type holders with modest positions, the upper edge a fully invested trader. In Bitcoin's early years the band was very wide, and Bitcoin returned far more than the middle of it: 219% a year on average over 2012–17, when the middle of the band was 110%. Since then the band has narrowed and Bitcoin's returns have come into line with it: 46% a year on average over 2018–26, when the middle of the band was 36%. The convergence was gradual, not a break at one date (Section 5). The test has limits (Section 5): fourteen years of a volatile asset measure an average only roughly, and the required return says nothing about the timing of booms and busts. On the same test the age rule does about as well. What separates the two is construction: the required return is built from risks rather than extrapolated from the price trend. On this reading the age rule is a reduced form of a required return that was itself falling; it now sits above the band.

What this means in practice. Bitcoin's price has to rise by about 18% in an average year to keep its current holders, as concentrated and as risk-averse as they are, willing to hold it. A buy-and-hold investor compounds at less than the average, because gains and losses of equal size do not cancel (a 50% loss needs a 100% gain to recover); at Bitcoin's volatility the drag is about 9 points, so 18% in an average year compounds at about 9% a year. The 18% is the figure to compare with other expected returns, the 9% the rate at which wealth would grow.

For an allocator three things follow. First, extrapolating Bitcoin's history (the 32%) overstates its

[2]The concentration term is risk aversion times variance times portfolio weight, $\gamma\sigma^2 w$ (Section 2). At $w = 1$, $\gamma = 1$ and 120% volatility it is $1.2^2 = 1.44$, or 144 points a year above the risk-free rate; the option-implied variance is a floor on this all-in requirement. Today, with γ of 1–2.8, $\sigma = 42\%$ and w of 8–19%, the same term is 1–9 points.

prospective return by more than two-fifths. Second, Bitcoin is not a free diversifier: about 7 of the 18 points are payment for stock-market risk the allocator already holds. An investor who is diversified and holds no Bitcoin yet would, on a first purchase, bear that market risk and a small risk of death but no concentration risk, so she should require less of Bitcoin than its present holders do: the 5% risk-free rate, plus 6–8 points for its beta, plus 1–2 for death, or 12–15%. (The S&P 500's own hurdle on the same inputs is 9%.) Third, the 18% tells her how much to hold. On her first dollar of Bitcoin she needs 12–15% and the market offers 18%, so she buys. But every dollar she adds makes her own portfolio more concentrated in Bitcoin, and the return she needs rises with it: by about half a point for each 1% of her wealth, for an investor of moderate risk aversion. She stops when the return she needs has climbed to the 18% on offer, which on the central numbers happens at about 11% of financial wealth (Section 8). A more cautious investor stops sooner; a believer in the 32% would hold more than three times as much.

Contributions and plan. The paper makes four contributions. The first is a normative required return for Bitcoin's actual holders with every input measured; the equation is set out in Section 2 and assembled for today in Section 4. The second is a set of mortality curves for cryptoassets built for pricing, with the Lindy hypothesis tested and rejected and the top-asset hazard measured (Section 3 and Appendix C). The third is a month-by-month test of the required return against Bitcoin's realized returns since 2012 (Section 5). The fourth is a dated, falsifiable forecast for September 2026–2031: about 18% a year on average (90% range 15–22%), about 9% a year compounded, against the age rule's 32% (Section 6). As checks, the same equation is applied to Ether and smaller cryptoassets (Section 7) and to gold (Appendix E), where it reproduces a known result. Section 8 translates the results for an allocator, Section 9 lists limitations, and Section 10 concludes and sets out the procedure as a template for other new assets. Appendices A to C take the three risk terms in turn, and Appendix D covers assembly and validation.

2. THE REQUIRED-RETURN EQUATION

Everything in this paper runs through one equation, so we build it slowly and then display it.

Expected versus required. An asset's *expected return* is a forecast: what it will earn on average. It is a statement about the price process that only realized returns can check, and they check it badly: Elton (1999) shows that realized returns are so noisy a proxy that decades of data can fail to distinguish competing values. A *required return* is a hurdle: what an investor needs the asset to earn to be willing to hold it, given its risk. What Bitcoin's holders actually require is not observed directly. It can be read from what holders say (Greenwood and Shleifer 2014), from the portfolios they choose (Giglio, Maggiori, Stroebel, and Utkus 2021), and from option prices (Martin 2017), or it can be derived from a model of what an investor ought to demand. The CAPM is the familiar model of that kind (Sharpe 1964; Lintner 1965), but its investor is fully diversified: a hurdle, not a description of anyone who holds Bitcoin. This paper derives the hurdle for the holder Bitcoin actually has, with her measured concentration, and checks it against what holders say and what options imply (Section 4) and against what the asset delivered (Section 5). The last check rests on equilibrium: the price adjusts until the expected return equals what its holders require. That equality is the whole content of "the market is in equilibrium"; it is why Cochrane (2011) can treat expected returns and discount rates as one object, and testing it is what empirical asset pricing does (Fama and French 2004). For an asset whose holders are still learning what it is, the gap can stay open for years, because learning makes realized returns predictable in ways no required return explains (Lewellen and Shanken 2002; Pástor

and Veronesi 2009a); the separation of the two returns for valuation is developed in Papadogiannis (2026a, 2026b).

Whose requirement? A required return is always someone's. A pension fund might be happy to hold Bitcoin at an expected 8%, but if pension funds do not hold it, their 8% has no bearing on the price. The price is set by the people who do hold the asset: it must offer enough to keep the least willing of them, the *marginal holder*, from selling. When this paper speaks of Bitcoin's required return it means exactly that: the return Bitcoin must offer, given its risks and who holds it today, for its holders to go on holding it. It is a property of the asset and its holder base, not of one investor, and it changes when the holders change, which is the source of the paper's forecast.

What risk aversion means. A risk-averse person turns down a gamble even when its odds are fair, because a dollar matters more when you have fewer of them: a loss takes away dollars you valued highly, and an equal gain adds dollars you value less. Risk aversion, γ (the Arrow–Pratt coefficient), measures the strength of this effect: when wealth rises by 1%, the value placed on one more dollar falls by γ%. It does not say that the rich are more cautious than the poor: the standard assumption is that γ is the same at every level of wealth ("constant relative risk aversion"), so two investors with the same γ turn down the same gamble when it is stated as a share of wealth. Estimates keep landing between 1 and 4: about 2 from US household asset holdings (Friend and Blume 1975), below roughly 2 from labor-supply behavior (Chetty 2006), a median of 1.7 in Italian household panels, which also support the constant-relative form (Chiappori and Paiella 2011), and small single digits in the macro calibrations that followed Mehra and Prescott (1985); 2–3 is the workhorse range.

What concentration costs. The consequence is that risk has a price: about $\gamma/2$ times the variance of the risk, both as shares of wealth. Someone with \$100,000 who must stake \$10,000 on the toss of a coin faces a variance of $0.10^2 = 0.01$ and, with $\gamma = 2$, would pay about \$1,000 to be excused (\$500 with $\gamma = 1$, the benchmark economists call log utility). Holding a volatile asset is the same kind of gamble. An investor with a fraction w of her wealth in an asset of annual volatility σ, for the moment uncorrelated with the rest of her portfolio (its comovement with equities is carried by the market term added below), bears a variance of $\sigma^2 w^2$, which costs her $\gamma\sigma^2 w^2/2$ a year. Should she hold a little more? The benefit is the asset's expected return above the risk-free rate, the *excess return*, since the dollar could otherwise sit in a Treasury bill. The cost is the rise in that risk charge from one more dollar, which is $\gamma\sigma^2 w$. She stops adding when the two are equal: expected excess return $= \gamma\sigma^2 w$. The three multipliers are symmetric: halving her weight lowers the return she requires exactly as much as halving her risk aversion or the variance would, which is why a dilution of the holder base, as in the ETF era, lowers the required return just as a fall in volatility does. With Bitcoin's current volatility $\sigma = 0.42$ ($\sigma^2 \approx 0.18$) and the surveyed share of crypto in holders' financial assets, $w = 0.19$ in 2021 (Section 4), a holder with $\gamma = 2$ requires $0.18 \times 0.19 \times 2 \approx 6.7$ points a year above the risk-free rate for the volatility alone; a holder with only 5% of her wealth in Bitcoin, 1.8 points. Section 8 reads this as a sizing rule. Here it is read the other way: an observed weight and an observed volatility reveal the excess return the holder must have been requiring. Adding the two risks this simplest version leaves out, market risk and sudden death, gives the central equation of the paper, which we call the **required-return equation**:

$$\mu - r_f \;=\; \underbrace{\beta\,\mathrm{ERP}}_{\text{market risk}} \;+\; \underbrace{\gamma\sigma^2 w}_{\text{concentration risk}} \;+\; \underbrace{\lambda\,(1-\delta)}_{\text{mortality risk}} \tag{1}$$

On the left is the asset's expected return over the risk-free rate; on the right are the three risks of the Introduction, each stated per dollar held in the asset. β is Bitcoin's beta to the equity market and ERP the equity risk premium. The **concentration term** is γ times variance times the holder's own portfolio weight, the risk she cannot diversify because she chooses, or is forced, to hold a lot of one thing. The **mortality term** needs a paragraph of its own.

The mortality term. A holder needs compensation for the risk that the asset dies, equal to the loss she can expect from it. That expected loss is the product of two numbers. λ is the probability per year that the asset dies. δ is the *recovery fraction*, the share of her money a holder keeps when it does, so $(1-\delta)$ is the share she loses. They are multiplied because dying and losing everything are not the same thing: when TerraUSD collapsed, holders kept about two cents on the dollar ($\delta \approx 0.02$); when BUSD was wound down under regulatory order, holders redeemed at par ($\delta \approx 1$) and nobody lost a cent. A 3% annual death probability with 95% lost on death is an expected loss of $0.03 \times 0.95 \approx 2.9$ points a year, and that is what the price must add, in the years the asset lives, to what holders otherwise require. A surviving asset's history contains this compensation in full, because the losses it was paid for never arrived (Brown, Goetzmann, and Ross 1995). The required return of equation (1) contains it too, which is what makes the two comparable.

Where the equation comes from. Each term of equation (1) rests on an established result, and that is deliberate: a required return built from accepted pieces can be checked piece by piece. The concentration term is the first-order condition of textbook portfolio choice (Merton 1969), and its use as a pricing statement under limited participation is Merton (1987): when only a fraction of investors hold an asset, those who do must be paid for the idiosyncratic variance they absorb, in proportion to how much each absorbs. The mortality term is the standard jump compensation (Merton 1976) read through the survival conditioning of Brown, Goetzmann, and Ross (1995), the form, hazard times loss, used in event-risk allocation (Liu, Longstaff, and Pan 2003; Das and Uppal 2004) and in the rare-disaster models that followed Barro (2006). The beta term is the CAPM, and the whole equation reduces to it when the hazard λ is zero and the holder's weight w shrinks to Bitcoin's share of all investable wealth, well under 1% today: it is the CAPM extended by the two risks the CAPM assumes away. What is new is the assembly and the measurement. Equation (1) joins the three terms into one pricing statement that nests the CAPM, and matches each to something observable for this asset class: the holder's weight to a survey, the death hazard to a census of the asset's relatives, the whole to stated expectations, option prices and fourteen years of returns. Required returns for Bitcoin have been read from option prices (Almeida, Grith, Miftachov, and Wang 2024) and from factor regressions (Liu and Tsyvinski 2021). To our knowledge none has been built from the holders' side, from their measured portfolios and a census of the asset class's deaths. The concentration mechanism has a name and an empirical record. It is Merton's (1987) *investor recognition hypothesis*: assets held by few carry a higher cost of capital, which falls as recognition broadens. Levy (1978) and Errunza and Losq (1985) derived the same form for investors who hold few securities and for securities that foreigners cannot hold. The hypothesis has been tested for individual stocks, where broadening the shareholder base raises the price and lowers subsequent returns (Kadlec and McConnell 1994; Foerster and Karolyi 1999; Amihud, Mendelson, and Uno 1999; Lehavy and Sloan 2008; Bodnaruk and Östberg 2009), and for national markets opening to foreign capital (Bekaert and Harvey 2000; Chari and Henry 2004). Pástor and Veronesi (2009b) give it a lifecycle: a new technology's risk is idiosyncratic while it is small and held by few, and turns systematic as it is adopted. Ibbotson, Idzorek, Kaplan, and Xiong (2018) generalize it into a theory of popularity, in

which hard-to-diversify assets earn a premium that popularity erodes. For cryptoassets the record is indirect: a broadening investor base is linked to idiosyncratic volatility (Izadyar and Zamani 2022), and institutional participation, the spot ETFs in particular, raised Bitcoin's correlation with equities (Copestake, Furceri, and Terracciano 2026), as index investment did for commodities (Tang and Xiong 2012). Every test above is a sign test on prices or correlations; this paper measures the level.

Four clarifications. First, on what the equation explains. A full model of Bitcoin's price would start from deeper things, such as the supply schedule, the pace of adoption and the costs of miners, and derive from them the price, its volatility, and who holds the asset (Biais, Bisière, Bouvard, Casamatta, and Menkveld 2023; Pagnotta 2022; Cong, Li, and Wang 2021). Such models explain the price but take the rate at which investors discount Bitcoin's future as given, or leave it to be estimated. That rate is the object of this paper. We take volatility, the holder's weight and the death hazard as they are observed and derive from them the return holders must require: the number that an allocator, a valuation, or one of those models needs, and that none of them supplies. Second, the equation is special neither to Bitcoin nor to one holder. The same line prices founder stock, private equity, and emerging-market equities before liberalization; Bitcoin enters through the values of the inputs, and through one data advantage: a thousand-asset graveyard from which λ can be measured rather than assumed. And every holder satisfies the equation with her own weight; the price settles where the holder who is only just willing to stay is satisfied, so in equilibrium her requirement is the return the market as a whole expects. Emerging markets show the mechanism our forecast rests on (Bekaert and Harvey 2000; Henry 2000): required returns are high while a market is segmented and fall when it opens to outside capital; the ETF era is Bitcoin's liberalization. Third, on using the survey average for w. If holders have similar risk aversion, each adds Bitcoin until $\gamma\sigma^2 w$ reaches the same premium, so their weights end up alike and the average holder's weight can stand in for that of the holder who sets the price. This assumes that holders chose their weights on beliefs that are right on average. A weight inflated by over-optimism satisfies the same equation but reveals a belief, not a fair premium, and only the comparison with realized returns in Section 5 can tell the two apart. Fourth, on whom the equation describes: an investor whose risk aversion does not change with her wealth, who faces two kinds of risk in Bitcoin, ordinary price swings and the chance of sudden death, and who keeps the rest of her wealth in the broad market (Appendix A).

What the equation leaves out. Equation (1) does not price everything, and what it leaves out has known signs. Pushing the required return up: a liquidity premium (Amihud and Mendelson 1986), material in Bitcoin's early years and small today; a jump-risk premium above the actuarial loss if deaths are systematic, which Section 8 charges for; ambiguity about the distribution itself, natural for a new asset; hedging demands against state variables such as monetary policy, to which Bitcoin has become sensitive (Copestake, Furceri, and Terracciano 2026). Pushing it down: a convenience or service yield from holding the asset, the analogue of gold's unobserved dividend in Barro and Misra (2016); a lottery preference for skewness (Barberis and Huang 2008); and disagreement under short-sale constraints, which raises prices and lowers expected returns (Miller 1977). The defense is not that these are zero but that two checks leave little room for a large net effect. First, option prices give the return that a holder with *all* her wealth in Bitcoin requires, whatever risks she is pricing: 21–35% a year today in nominal terms, the 5% risk-free rate plus 16–30 points (Section 4). A holder with only 8–19% of her wealth in Bitcoin bears far less of its risk and must require less than that. Our 18% for her does. Second, delivered returns have matched the required return within about ten points a year

since 2018 (Section 5); the early years, when they ran above its midpoint, are where an omitted premium would show.

What moves each term. Volatility falls as markets deepen (Bitcoin's from about 150% in 2013 to 40–50% in 2026). The weight w falls as participation broadens: when the only people willing to hold an asset are enthusiasts, each must hold a lot of it; when ETFs let everyone hold a sliver, nobody has to. The market term moves the other way: integration with mainstream portfolios raises the comovement with equities. The mortality term is the one with no trend. The census finds that the hazard of cryptoassets does not decline with age (Section 3), so we hold it constant, and new threats such as quantum computing could raise it (Appendix C). Figure 1 in Section 5 shows the path of each.

3. DATA, AND THE CENSUS OF CRYPTOASSET DEATHS

Each term of equation (1) has its own data. This section says where each comes from; the appendices give the detail.

Market risk and the risk-free rate. Beta is estimated on month-end total returns of the S&P 500, represented by the Vanguard S&P 500 ETF (Yahoo Finance), over four-year windows, and priced at a constant 4% equity premium; Appendix A discusses both choices. The risk-free rate is the ten-year Treasury yield. The equity comparison of Table 6 adds the iShares MSCI ACWI ETF for global equities. The gold comparison of Appendix E uses LBMA gold prices and World Gold Council above-ground stock estimates.

Concentration risk: prices, surveys and options. Volatility comes from Bitcoin's monthly prices (Coin Metrics 2026). The holder weight comes from household surveys: crypto-ownership rates from the Federal Reserve's payment and household surveys and Pew (0.4% of US adults in 2014, 1.2% in 2018, 11% in 2021, 12% since), and owners' portfolio shares and expected returns from Candia, Coibion, Gorodnichenko, and Weber (2026), with Austrian household-finance data (OeNB 2025) as a European cross-check. Carrying the weight through time also uses Bitcoin's market capitalization and address counts (Coin Metrics), global household wealth (UBS Global Wealth Report), and spot-ETF and public-company holdings from issuer disclosures and corporate filings (as compiled by Bitcoin Treasuries 2026); Appendix B shows each step. Risk aversion comes from the household-finance literature and from surveys of crypto owners' risk tolerance (Section 4). The requirement of a fully invested holder comes from Deribit option prices, as reported in the literature, and the DVOL index.

Mortality risk: a census of cryptoassets. The asset panel is the Coin Metrics community dataset, observed from July 2010 to May 24, 2026. After removing fiat exchange rates, tokenized equities, wrapped and staked duplicates, stablecoins, and trading-pair artifacts, the universe is **1,815 cryptoassets** and about 67,700 asset-months of exposure. Daily trading volume is available for essentially all of them and daily prices for the 139 historically significant ones, so volume carries the life-and-death timeline and prices the return calculations. An asset is born on the first day it shows a price or volume. It is dead when its data feed ends more than 180 days before the dataset does (234 cases), when its price falls below 1% of its all-time high and never recovers (33 cases, on the price-complete subset), or when its trailing three-month volume falls below 1% of its peak and below $250,000 a day and never recovers (438 cases, overlapping the other two; the absolute floor keeps healthy mid-caps whose volume merely normalized after 2021 from being counted). Two corrections are made by hand, with the audit trail in Appendix C. Ticker migrations are not deaths: when MATIC became POL or EOS became Vaulta, holders received successor tokens at full value,

and seventeen such events are removed. And known catastrophes are dated from the public record: LUNA and UST at May 12, 2022 and FTT at November 10, 2022. The result is **520 deaths among 1,815 assets (29%)**, of which 8 are sudden, value-destroying *jump-collapses* and the rest gradual abandonment. One bias should be stated up front. Coin Metrics adds an asset only once it has become significant, so micro-caps that died young never enter the panel. This *understates* the death rate at young ages, which works *against* our finding that hazard rises with age. The elite estimates that Bitcoin's pricing needs are unaffected: an asset in the top twenty is fully observed from then on.

What the census shows. Two results matter for pricing; Appendix C has the tables, figures and tests behind them. First, cryptoassets do not become safer with age. The popular "Lindy" idea, that the longer an asset has survived the longer it will go on surviving, would make the death rate fall like one over age, and could have explained Bitcoin's history as fair payment for a risk of death that fades. The data reject it. The death rate rises from under 5% a year in an asset's first year to 12–17% between ages three and nine, and a regression of deaths on age with calendar-year effects gives an elasticity of **+0.42 (s.e. 0.04)** where Lindy requires -1. Cryptoassets age as young firms do, whose risk of failure also rises over the first years of life before it falls (Brüderl and Schüssler 1990), and not as the Lindy idea supposes; the wider parallel between cryptoassets and firms is drawn in Papadogiannis (2025, 2026d). Second, size is what protects (Table 1): the death rate falls from 17.4% a year in the smallest quartile by trading volume to 3.2% in the largest. The last two lines of the table are the ones Bitcoin's pricing needs. Among the 219 assets that ever reached the top twenty by monthly volume, 36 died over 1,017 asset-years of exposure, an all-cause rate of 3.5% a year, and only three of those deaths were the sudden, catastrophic kind that a jump-risk premium prices: LUNA and UST in May 2022 and FTT in November 2022, each a roughly total loss within days. None has occurred since. The other elite deaths were slow fades, in which holders could exit at declining prices; losses of that kind live in an asset's volatility and average return, not in its jump hazard.

Table 1: Mortality by size and for elite cohorts (curated deaths).

Size group	n	Hazard (per yr)
Smallest quartile	452	17.4%
Q2	451	13.0%
Q3	451	9.8%
Largest quartile	452	3.2%
Ever top-50 by volume	499	5.5% (CI 4.5–6.5%)
Ever top-20 by volume	**219**	**3.5% (CI 2.5–4.9%)**
Ever top-20, jump-collapse only	219	**0.30% (CI 0.06–0.86%)**

What this gives the pricing equation. The 0.30% is the probability λ; the loss is nearly total. Holders of the three jump deaths kept between one and five cents on the dollar (Appendix C.3), so we set the share lost, $1-\delta$, at 0.95: $\lambda(1-\delta) \approx 0.3\% \times 0.95 \approx$ **0.3 points a year**. Treat every elite death, even the slow fades, as a jump at total loss and the term is still only about **3 points** (3.5 with every death, 2.4 without the 2025 delisting wave). The mortality term in equation (1) is real (2022 proved it can strike the top ten), but it is a few points. Two caveats. No cross-section can measure Bitcoin's *own* hazard in 2010–2014, when it was the only asset of its kind and plausibly faced existential risk; our bounds apply from the point at which an elite class exists to measure, and Section 5 keeps that early window wide. And risks that have never yet fired for anyone, the clearest being a quantum break of Bitcoin's signature scheme, are covered by the

equation but cannot be priced from the census (Appendix C).

4. WHAT BITCOIN'S HOLDERS REQUIRE

The required-return equation, equation (1), needs five inputs. The graveyard supplied the mortality term. This section supplies the holder's portfolio weight and risk aversion, takes volatility from prices and the market beta from Appendix A, assembles the required return for 2026, and sets it against two outside readings, holders' stated expectations and option prices.

Source 1: holders' measured portfolios. The Cleveland Fed study of cryptocurrencies in household finance (Candia, Coibion, Gorodnichenko, and Weber 2026) reports the number the equation needs most: among US households that owned any crypto in 2021Q3, crypto made up **19% of financial assets on average**, with 40% of owners below 5% and about a fifth of owner-households above *half* their portfolio. A European cross-check points the same way: in the Austrian central bank's 2023 household survey, owners mostly held under 20% of financial assets in crypto, rarely above 20–25% (OeNB 2025). So the typical holder of the early 2020s was not the diversified investor of the CAPM; she held a fifth of her wealth in the asset class. The surveys measure crypto as a whole, not Bitcoin alone, and we use the crypto share as Bitcoin's w. This is the quantity the equation calls for. The concentration term prices how much of the holder's portfolio moves with Bitcoin, which is her Bitcoin share plus the share of each other coin times that coin's beta to Bitcoin. Those betas are 0.85–1.15 (monthly, since 2018, for Ether and five other large coins), so the sum is close to her crypto share. The 2025 wave of the same survey reports portfolio shares similar to those of 2021 without publishing the figure. For 2026 the paper therefore uses a range: 19% as the upper bound (for survey-type holders the market-clearing back-cast of Section 5 returns 23%, and 15% when it is run on all cryptoassets, Appendix D) and 8% as the lower bound, obtained by adding the spot-ETF and public-company treasury holdings, about 2.5 million BTC by mid-2026, as a second class of holders who are not "crypto owners" in any survey and hold at about 1.5% of their wealth (Appendix B).

Source 2: holders' stated expectations, a partial check. The same surveys asked holders what return they expected over the next year: 22% in 2021Q3, 13.8% in 2025Q2. Table 2 sets each answer against equation (1) evaluated with the inputs of its date.

Table 2: Two survey waves, one equation.

	2021Q3	2025Q2
Volatility σ (trailing, at the survey date)	77%	53%
Holder portfolio weight w	0.19 (measured)	0.19 ("similar across waves")
Risk-free rate	1.5%	4.4%
Market term (rolling beta $\times$ 4%)	5.2 pts ($\beta = 1.3$)	7.8 pts ($\beta = 2.0$)
Required return at $\gamma \in [1, 2.8]$	18–38%	18–27%
Stated expected return	22%	13.8%
γ that reproduces the stated return	≈ 1.3	≈ 0.2

Mortality term at 0.3 points. Volatility is the paper's trailing estimate averaged over the quarter each wave was fielded; Candia et al. report the 2025 wave as 2025Q2 and the 2025 portfolio share as similar to 2021's. Betas are the rolling four-year estimates of Appendix A at each date.

The table is not a clean confirmation, and we report it as it is. In 2021 holders' stated 22% lies inside the model's 18–38% and is reproduced at a risk aversion of about 1.3, close to log utility: beliefs and positions

cohere, as Giglio, Maggiori, Stroebel, and Utkus (2021) find for equity investors. In 2025 they do not. Holders said 13.8%, barely more than the risk-free rate plus the market term alone (12.2%), which leaves almost nothing for a fifth of their wealth in an asset with 53% volatility; it would take a risk aversion of about 0.2. Either holders in 2025 expected less than their own risk required, or they do not perceive the market risk the rolling beta measures, or a twelve-month point forecast given after a strong run is not the long-run expectation the equation prices. The paper therefore does not calibrate risk aversion to the surveys.

The range for risk aversion. The lower end of the range for γ is below the household estimates of Section 2, because the people who choose to hold crypto are more tolerant of risk than other households. In the Federal Reserve's 2022 household survey about 40% of crypto investors rated their willingness to take financial risk at 7 or more out of 10, against under 20% of non-owners (Hayashi and Routh 2025); Austrian owners are likewise more risk-tolerant than non-owners (Stix 2021), and retail investors who buy crypto products hold risky portfolios and tilt further toward risk after their first crypto purchase (Hackethal, Hanspal, Lammer, and Rink 2022). These studies rank holders against non-holders but do not measure γ. A lower bound of 1, log utility, is our reading of them; it brackets the 1.3 implied by the 2021 survey. The choice matters. With the lower bound at 1.7, the median household of Chiappori and Paiella (2011), the central estimate below would be 18.9% instead of 18.0%, and as γ goes to zero the required return falls to 12–15%, the diversified investor's hurdle of Section 8, which is where the 2025 answer sits.

Source 3: option prices. Options answer a question the surveys cannot: what would a holder with *everything* in Bitcoin require? No holder can be more concentrated, so her requirement is a ceiling for the concentration term. Martin (2017) shows that it can be read from option prices: an investor fully invested in an asset, with log utility ($\gamma w = 1$ in equation (1)), requires an excess return equal to the asset's variance as option prices imply it, not as history shows it. For July 2017 to December 2022, when implied volatility ran at 80–90%, Deribit options put this at **66–84 points a year** (Almeida, Grith, Miftachov, and Wang 2024). Crypto markets were then segmented from mainstream capital, with large and persistent price gaps across exchanges (Makarov and Schoar 2020), so price-setting traders close to all-in are plausible for those years, and Section 5 uses this figure as the upper edge of the required band. Today implied volatility (Deribit's DVOL index) has run between about 40 and 55, so the same all-in holder would require 0.40^2–$0.55^2 \approx$ **16–30 points of excess return**, or 21–35% a year in nominal terms. Today's holder is not all-in: with w of 8–19% she requires **1–9 points** for concentration, on top of the risk-free rate and the market term. Options give the ceiling; the surveys say how far below it holders now sit.

Putting the 2026 number together. Table 3 lists every input of equation (1) and says where each end of its range comes from. On a risk-free rate of 5.0%, the market term is 6–8 points, concentration 1–9 points and mortality 0.3–3 points. Summing the extremes gives 12.7% to 25.4% a year, an *envelope* of 13–25%. But an envelope puts every uncertain input at its extreme at once, and the inputs are independent. Treating each as uniformly uncertain over its stated range and drawing them independently (Appendix D), the required return has a median of **about 18% a year**, a central 68% interval of 16–20%, and a **90% interval of 15–22%**. The market term is now the largest single component (about 7 points), followed by concentration (about 4.5); the holder weight still contributes most of the width. This is an arithmetic average; the compound return it implies is lower by about half the variance, roughly 9 points at today's volatility, so about 9% a year. The realized five-year outcome will carry, on top of this, noise of roughly ± 19 points on the compound return at one standard deviation, which no forecast of a 42%-volatility asset can remove.

Table 3: The inputs to the 2026 required return, and where each range comes from.

Input	Value	Low end	High end
Risk-free rate	5.0%	One value: the ten-year Treasury yield in 2026.	
Beta	1.5–2.0	1.5, estimated on all months since 2016.	2.0, the upper part of the four-year estimates since 2022 (1.6–2.2); the latest four- and five-year estimates are 1.75 and 1.8.
Equity premium	4%	One value, forward-looking (Fama and French 2002); 6% is carried as a high case (Appendix A).	
Risk aversion, γ	1–2.8	1 (log utility): crypto owners tolerate more risk than other households, and the 2021 survey implies 1.3.	2.8, in the upper part of the 2–3 range common in household finance; three-quarters of households are below 3 (Chiappori and Paiella 2011).
Volatility, σ	42%	One value: the volatility of monthly returns in May 2026, exponentially weighted with a half-life of nine months. Option-implied volatility has run at 40–55%; Appendix D varies σ.	
Holder weight, w	8–19%	8%: surveyed owners together with ETF and corporate-treasury investors, who hold about 1.5% of their wealth in Bitcoin.	19%: the crypto share of owners' financial assets in the Cleveland Fed survey (2021; similar in 2025).
Mortality, $\lambda(1-\delta)$	0.3–3 points	0.3: sudden collapses only, 0.30% a year at a 95% loss.	3: every death of a top-twenty asset, slow fades included, counted as a total loss (2.4–3.5% a year).

Market term: beta times the equity premium. Concentration term: $\gamma\sigma^2 w$.

5. THE REQUIRED RETURN THROUGH TIME, AGAINST WHAT BITCOIN DELIVERED

A cost of capital earns its name if, over time, the asset delivers about what it says. This section runs that test: the required return is evaluated at each date from 2012 from that date's inputs and compared with what Bitcoin then delivered. The start is set by the data: Bitcoin's price series begins in July 2010, and January 2012 is the first month with enough history to estimate its volatility.

Matching units. The comparison has to be made in the units of equation (1). The required return is an arithmetic average: the return in an average year, the number asset-pricing equations quote as an expected return. It is not what a buy-and-hold investor compounds at. An asset that gains 100% one year and loses 50% the next has an average return of 25% a year and a compound return of zero; the difference is the drag of volatility. In general the compound annual growth rate (CAGR) is lower than the average by about half the variance, and at Bitcoin's historical volatility of 60–150% that is 20 to 100 points a year. Comparing a required average return with a realized CAGR would understate what Bitcoin delivered by that much. Two realized measures are used below, both arithmetic averages. The test uses monthly returns: twelve times the average simple monthly return is a direct estimate of the average return and needs no correction. Figure 1 shows the path smoothed over five-year windows, where the *realized average return* is the mean log return plus half the realized variance, annualized, with the CAGR alongside for reference.

The required return at each date, $\mu^{req}(t)$, is equation (1) evaluated from the ingredients measured at that date: trailing volatility from the price series; the risk-free rate of the date; the market term from a rolling four-year beta (Appendix A); the mortality term held at its stated range throughout, since the elite hazard is

one average over 2014–2026; and the holder weight w, which is not observed before 2021 and is back-cast by market clearing (Appendix B). Bitcoin's capitalization must equal the wealth its holders have placed in it, so the average weight is capitalization divided by participating wealth. We anchor participating wealth in 2021Q3, the one date where both sides are measured, and move it through time with the surveyed share of adults holding crypto (0.4% in 2014, 1.2% in 2018, 11% in 2021Q3, 12% since) and a ramp in wealth per participant; from 2024 the diluted reading also counts the roughly 2.5 million BTC in spot ETFs and public-company treasuries as held at 1.5% weights. The implied weight is about 3% in 2013, 17% in 2017, 18% in 2021, and in 2026 between 8% with the institutional channel and 23% without it. The required return is a band because several inputs are ranges, and its two edges differ in all of them, not in γ alone. The lower edge takes the low end of each: $\gamma = 1$, the diluted weight, mortality of 0.3 points, and a market term 15% below its estimate. The upper edge takes the high end ($\gamma = 2.8$, the weight without the institutional channel, mortality of 3 points, a market term 15% above) or, where that is larger, the requirement of the all-in trader of the option bound, which sets the upper edge in most months before 2023 and in none since (Appendix D.3). The midpoint is the average of the two edges. Panel (a) of Figure 1 shows the band month by month, and Table 10 in Appendix D tabulates it at two-year intervals.

The test. Equation (1) is a condition on the expected return at each date, so the direct test pairs the required return at each month-end with Bitcoin's return over the following month. That gives 172 non-overlapping observations, January 2012 to April 2026 (Table 4). Over the full sample Bitcoin delivered an average of 118% a year. The required band averaged 22–116%, with midpoint 67%. The midpoint falls short of delivery by 51 points a year with a standard error of 39: not statistically different from zero, and not a tight fit. The two edges separate. The lower edge alone (survey-type holders with modest weights) leaves 97 points unexplained and is rejected ($t = 2.5$); the upper edge (the all-in holder) leaves 2. The data therefore favor the upper half of the band in the early years; the midpoint is a convention, not an estimate. A caution: a single month, November 2013, when the price rose 451%, contributes 31 of the 118 points, and without calendar 2013 the gap to the midpoint is 8 points (standard error 21).

Table 4: The monthly test: delivered against required return, percent per year.

			Required		Delivered minus required			Delivered minus
Period	Months	Delivered	Band	Midpoint	Lower edge	Midpoint (s.e.)	Upper edge	age rule
2012–26, all	172	118	22–116	67	+97	+51 (39)	+2	+48
2012–14	36	278	32–298	158	+246	+120 (169)	−21	+143
2015–17	36	160	22–105	63	+138	+97 (46)	+55	+82
2018–20	36	69	22–81	51	+47	+18 (48)	−12	+15
2021–23	36	30	16–42	29	+14	+1 (39)	−12	−12
2024–26	28	36	15–38	26	+22	+10 (33)	−2	+1
2012–17	72	219	27–202	110	+192	+108 (87)	+17	+112
2018–26	100	46	18–55	36	+28	+10 (24)	−9	+1

Each observation pairs the required return at a month-end, from that date's inputs, with Bitcoin's simple return over the following month; all figures are twelve times the monthly average. Delivered minus required is the average monthly difference, with its standard error in parentheses for the midpoint. Age rule: the one-month form of the rule, with the exponent held at its 2026 value, 5.62.

How close, and is it getting closer. Figure 2 shows the same test by three-year period. The gap to the

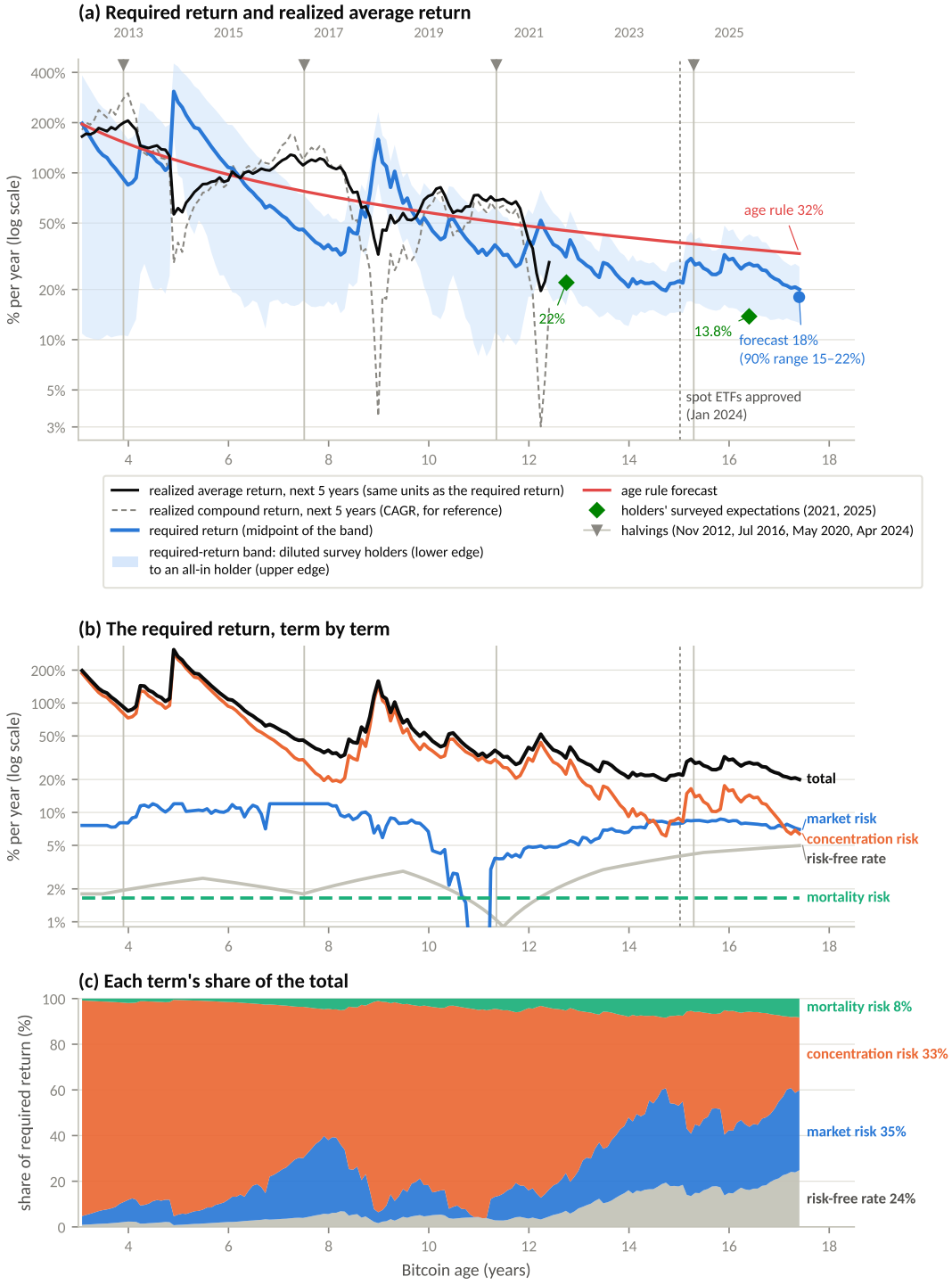


Figure 1: The required return since 2012, on one time axis. (a) The required-return band and its midpoint against Bitcoin's realized average annual return over the following five years (mean log return plus half the variance, the units of the required return) and the compound return over the same window; the realized curves end at age 12.4. Green diamonds mark holders' surveyed expectations, the dotted line spot-ETF approval, the triangles the four halvings; the right edge marks the 18% forecast against the age rule's 32%. Five-year averaging makes the path visible; the test is the monthly comparison of Table 4 and Figure 2. (b) The midpoint split into the four terms of equation (1); mortality is held at 1.65 points, and concentration is the remainder. (c) Each term's share of the midpoint.

midpoint is +120 points a year in 2012–14 (standard error 169) and +97 in 2015–17 (46), the one period in which the midpoint is rejected at conventional levels. In the three periods since 2018 it is +18, +1 and +10, each well inside one standard error. Over 2018–26 as a whole Bitcoin delivered 46% a year against 36% required, a gap of 10 points. Two things closed the gap. The band narrowed from about 270 points wide to about 20 as the two readings of who holds Bitcoin converged, so the midpoint stopped being a compromise between distant alternatives. And volatility fell, so each year of returns measures the average more sharply. What the test cannot do: a standard error of 24 points since 2018 cannot tell a 36% requirement from a 26% or a 46% one, and it cannot separate the required return from the age rule, whose gap is 48 points over the full sample and 1 since 2018. Nothing in fourteen years of returns contradicts the required return, and the fit is closest where its inputs are best measured.

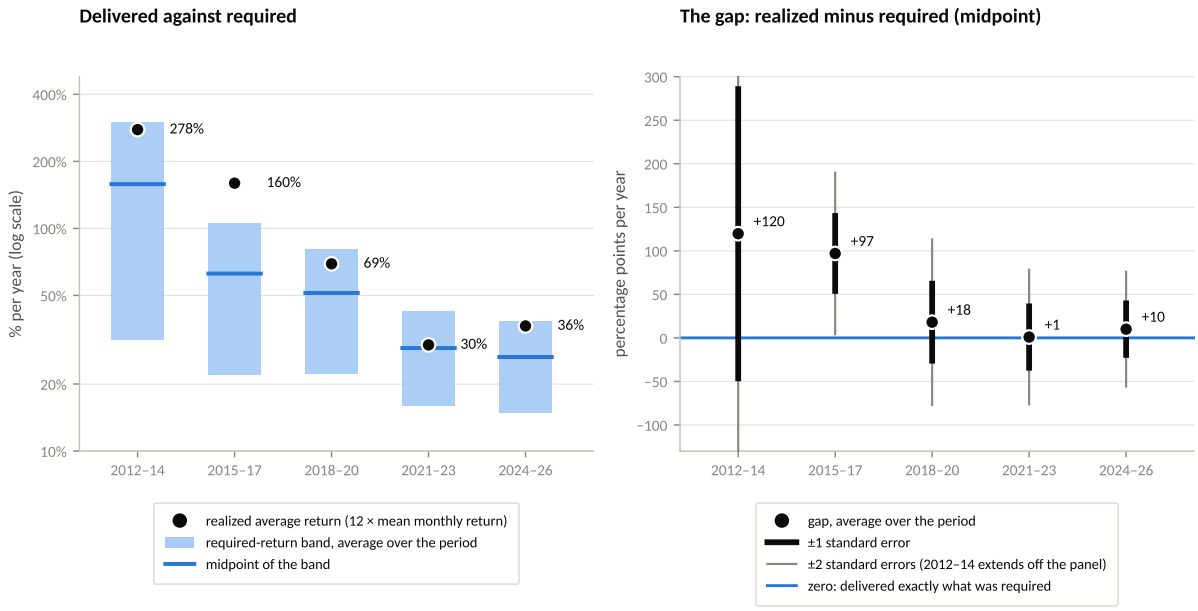


Figure 2: The monthly test by three-year period. Left: Bitcoin's realized average return (twelve times the mean monthly return) against the required-return band and its midpoint, averaged over the months of each period; logarithmic scale. Right: the gap, realized minus the midpoint of the band, with one and two standard errors. The 2024–26 period ends in April 2026.

What the test does not show. The test is one of level, not of timing, and on timing the required return has nothing to offer. Month by month its correlation with the following return is −0.01 (0.17 for the age rule). Over five-year windows the age rule fits the path far better (Appendix D). The reason is visible in Figure 1: the band is built from trailing volatility and from a holder weight with market capitalization in its numerator, so it is highest at market peaks, exactly when subsequent returns are lowest (Appendix D reports the regression). A date-by-date equilibrium restriction is not a timing signal, and the paper does not offer it as one. The age rule tracks the path better because it is a smooth trend extrapolated from the same price history; what it lacks is a reason, and so a way to know when it will stop.

Why the bands are wide early. The uncertainty in 2013–2017 is *who* was setting the price. Under the survey-anchored back-cast, participants were few but each held a modest weight, and the required return was about 20%. Under the all-in reading, which the 2017–22 option bounds support for the segmented era,

required returns were in the tens-to-hundreds. Delivered returns sit between the two readings, nearer the upper one in 2012–17 (Table 4), and for the paper's purpose the question does not need to be settled: on *either* reading the early engine (concentration, volatility, segmentation) has measurably shut down.

Which term carried the requirement, and when. Panels (b) and (c) of Figure 1 split the midpoint of the band into the four pieces of equation (1); read against panel (a), they show that each spike in the required return is a spike in the concentration term. In 2013 the concentration term, $\gamma\sigma^2 w$, was about 87% of a requirement near 126% a year. It still supplied about 70% in 2021. After the ETF-era dilution of the holder base and the fall in volatility the composition changed: in 2026 the market term is about 35% of the midpoint requirement, concentration 33%, the risk-free rate 24%, and mortality 8%. Two things follow. The history of Bitcoin's required return is mostly the history of one term. And the asset has changed character: its required return is now mostly the price of ordinary market risk and the risk-free rate, the profile of a high-beta risk asset, with a concentration premium that is still material but no longer dominant. (The midpoint path ends at 20% rather than the 18% of Section 4 because the midpoint weights the band's corners equally, which the propagation of Appendix D does not.)

What the age rule is, on this account. Both the required return and the age rule fell from over 100% to tens of percent across the sample, and a one-parameter power law in age is an excellent smoother of a declining curve. The age rule is best read as a reduced form of the required return: right on average over a period in which the inputs fell roughly as one over age, with no structure to say whether they will keep doing so. They have not. Volatility and the holder weight do not decline with age as such; they declined with integration, and the market term has risen with it. The age rule's 33% for mid-2026 (32% from September) sits above the top of the required band (27%), and the age rule's own out-of-sample record (Papadogiannis 2026c) is in the same direction: in windows beginning after the power law's 2019 publication, Bitcoin fell short of the age rule by 5–8 points a year.

Premium or belief? The concentration term is mechanical: a holder with a large fraction of her wealth in one volatile asset bears a great deal of undiversified risk, and $\gamma\sigma^2 w$ is the return that makes keeping it rational, whatever put her there. In Merton's (1987) story only a few investors can or will hold the asset, so each holds a lot and the price must pay them a premium, which arrives on average. In the other story investors pile in because they expect a great deal; their weights are the sizing rule of Section 8 run on inflated beliefs, and because the price has already risen to meet the belief, the return does not arrive. From the weight alone the two are indistinguishable. What separates them is whether the return arrived, and it did: delivered returns ran above the midpoint requirement before 2018 and have matched it since (Table 4). That is evidence for the premium reading over the sample as a whole, though not at every date: the required return was highest at the 2013, 2017 and 2021 peaks, and the returns that followed those dates were below it, which is what belief-inflated weights at peaks would produce.

The cross-section agrees from the side. If required returns rise with volatility, concentration and death risk, the typical altcoin (more volatile, more concentrated, higher λ than Bitcoin) required more than Bitcoin did. It delivered less than nothing in compound terms: across the significant-asset cohorts, the *median* coin born 2015–2020 shows a negative compound return at every horizon, while cohort *means* stay large on the back of a few winners, exactly the lottery shape that a 10–17% annual death rate plus a handful of extreme survivors produces. Within Bitcoin's own 2010–14 birth cohort at age ten, the mean coin compounded at about +167% a year, the median at +50%, and Bitcoin at +233%. Bitcoin is the asset for which realized

and required coincide on average; it is also the winner of its cohort, and a required return validated on the winner is validated on a selected path, a caveat Section 9 repeats.

6. THE FORECAST, AND WHAT WOULD PROVE IT WRONG

A required return is a statement about risk; a forecast is a statement about the price process. The bridge from one to the other is an equilibrium condition, which we state as our hypothesis: *Bitcoin is priced in equilibrium, at the return its holders require*, with no systematic mispricing in it. Section 5 gives the hypothesis some standing, since fourteen years of returns do not reject the required return and since 2018 it has matched delivered returns within about ten points a year, and the spot ETFs, the dilution of the holder base, and the compression of the option bounds all point to a market more integrated now than at any earlier date. Under that hypothesis, and only under it, the expected return equals the required return of equation (1), every ingredient of which is observable:

If Bitcoin is priced in equilibrium, its expected return for September 2026 to September 2031 is about 18% a year on average, with a 90% range of 15–22% (full envelope 13–25%); in compound terms about 9% a year (6–13%; envelope 4–16%). The age rule says 32% (22–28% compounded; see note 1). The two forecasts differ enough to be separated by a single five-year window, and we commit to ours.

Here is what each outcome would mean:

- **BTC compounds (CAGR) at $\geq\sim$25% a year to 2031.** The age rule is vindicated and this paper's account is wrong in an instructive way: either a required-return component we measure as small is in fact large (which the options market would have to show), or Bitcoin remains systematically underpriced, which is the age rule's implicit claim.

- **BTC compounds between the mid single digits and the high teens.** The account stands; the age rule's first structural break is confirmed, for the reason stated here.

- **BTC compounds near zero or below.** Within the band's logic this is a bad draw of the (large) diffusive risk, not a refutation. But if it comes with *rising* option-implied crash intensity well above the graveyard's 0.3–3%, the mortality term was underestimated.

Two notes on the test itself. *Units:* the 18% is a nominal US-dollar return and an arithmetic average; the realized outcome should be graded in the same units as the forecast it is compared with, as Section 5 does. *Asymmetry:* as the three outcomes make explicit, the forecast is sharply falsifiable upward but only conditionally falsifiable downward. A one-and-a-half-sided test is the price of forecasting a 45%-volatility asset, and we state it rather than hide it.

What about the halving cycle? Practitioners ask where the four-year "halving cycle," the peaks of 2013, 2017, and 2021 that followed the supply halvings of 2012, 2016, and 2020, fits in this account. First, the halving changes supply, and supply does not appear in equation (1): cutting issuance from 0.8% to 0.4% of the outstanding stock a year alters none of the risks a holder is paid for, so the required return is unaffected by the schedule. Second, the schedule is written in the code and known years ahead, and in an equilibrium a known future supply change is in today's price and cannot carry a predictable excess return around its date. A reliable, tradeable cycle would therefore be a systematic mispricing, and the equilibrium hypothesis

above says there is none left. Third, the past cycles were real and are in Figure 1, where the halvings are marked: the five-year windows starting at the 2013, 2017, and 2021 peaks are the troughs of the realized curve, those starting in the 2015 and 2019 troughs its crests. In this paper's language the cycle is realized returns swinging around the required return, and the equation says nothing about its timing, only that the swings average out in equilibrium, as they have since 2018 (Table 4). Fourth, the paper predicts that the cycle's amplitude should fade, because the mechanism that produced it, a small concentrated holder base at 80–150% volatility, is compressing (Figure 3), and because an integrated market arbitrages a four-year calendar. The forecast window, September 2026 to 2031, straddles the 2028 halving and whatever peak or trough follows it, so the cycle phase is partly averaged out of the test.

What about the upside stories? Practitioners also ask where the arguments for much higher prices fit: tokenization, payments by AI agents, a flight from government debt under fiscal dominance. These are claims about future demand, and the equation prices risk. A demand story that investors already believe is in today's price: it raised the price when it became believed and does not raise the return from here. If the market underestimates such a force and it arrives, Bitcoin will beat the forecast, and the gain accrues once, to those who held before the repricing. Afterward the asset is dearer and more widely held, and its required return is lower, not higher: broader ownership shrinks the concentration term, and an asset valued as a hedge is one that investors accept a lower return to hold, as with gold (Appendix E). Fiscal dominance changes the nominal figure only through the risk-free rate, which rises if yields follow inflation and does not if they are held below it; in neither case do the risk terms grow.

Two auxiliary predictions sharpen the mechanism. First, the required return should fall fastest where volatility and holder concentration fall fastest: Ether post-ETF should show the same compression; a still-segmented, still-concentrated large asset should not. Figure 3 shows the first check already passes: Ether's volatility and forward returns compress alongside Bitcoin's into the ETF era. Second, option-implied crash intensity should remain in the low single digits, consistent with the measured elite jump hazard; a reading persistently above $\sim$10% a year would push our band up and signal that the market disagrees with the graveyard, for instance about a quantum break of Bitcoin's signature scheme (Appendix C.6).

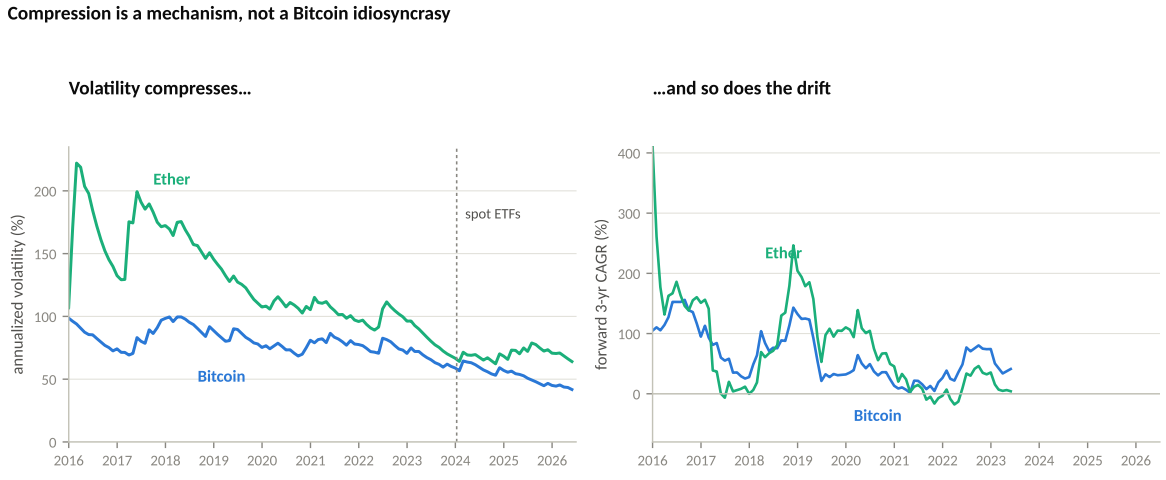


Figure 3: The compression mechanism is not Bitcoin-specific. Left: EWMA annualized volatility for Bitcoin and Ether, with spot-ETF approval dates marked. Right: realized forward three-year returns for both assets.

7. THE SAME EQUATION FOR ETHER AND OTHER CRYPTOASSETS

Nothing in equation (1) is specific to Bitcoin. It prices any asset held by concentrated investors who face a risk of total loss, and it does so by the same four steps: measure volatility and beta, find the holders' weight, count the failures of similar assets, and add the three terms to the risk-free rate. What changes from one cryptoasset to the next is the value of the inputs. The graveyard was built from the whole asset class, so it already supplies the death-risk input by size class, and Bitcoin's volatility and holder-weight measurements have natural counterparts. Table 5 walks the equation down the size ladder.

Table 5: The required-return equation across the cryptoasset size ladder. Required returns per year, as arithmetic averages. The central estimate and 90% interval come from drawing each input independently and uniformly over its stated range (Appendix D); the envelope sums the extremes. Risk-free 5%, equity premium 4%, $\gamma \in [1, 2.8]$ throughout. Beta is 1.5–2.0 for Bitcoin and 2.0–2.7 for Ether (since-2016 and four-year estimates); Ether's range is assumed for the altcoins. Altcoin holder weights are illustrative, not surveyed; small-coin volatility is taken as 100–130%.

	Bitcoin	Ether	Large altcoin	Small coin
Volatility σ, 2026	42%	64%	∼80%	100–130%
Market beta	1.5–2.0	2.0–2.7	2.0–2.7	2.0–2.7
Holder weight w	0.08–0.19	0.08–0.19	0.10–0.30	0.10–0.30
Death hazard, per year	0.3% jump	0.3% jump	3.2% all-cause	17.4% all-cause
Mortality term	0.3–3 pts	0.3–3 pts	1–3 pts	16–17 pts
Required return, central	**18%**	**26%**	**39%**	**78%**
90% interval	15–22%	21–34%	27–60%	52–124%
envelope	13–25%	17–41%	20–73%	39–175%
Delivered, average (compound)	27% (13%)	19% (−7%)	∼25% (−7%)	not observable

Delivered: Bitcoin and Ether, realized average and compound annual return from the 2021Q3 survey date to May 2026 (Section 5); large altcoins, the median compound return of the significant-asset cohorts born 2015–2020, measured five to seven years after birth (−6.5% to −8.7% a year), with the average obtained by adding half the variance at 80% volatility. Small coins have no usable price history in the data; what is observable is that half of the smallest quartile is dead within 3.3 years (Figure 8). Hazards are the graveyard's size-quartile and elite estimates (Section 3).

Ether. Ether has been a top-five asset for a decade, so its death risk sits in the same elite class as Bitcoin's and the mortality term is the same 0.3–3 points. It differs in two inputs: volatility, 64% against 42% today, and market beta, 2.0–2.7 against 1.5–2.0. On the holder weight we make one assumption: surveys report a household's crypto holdings as a block, Ether's monthly returns have a correlation with Bitcoin's of about 0.8, and so the share that matters for a holder's risk is her crypto share, the same 8–19%. The concentration term scales with the ratio of variances, $(0.64/0.42)^2 \approx 2.3$, and Ether's required return comes out at about 26% a year (90% interval 21–34%; envelope 17–41%). From the 2021Q3 survey date to May 2026 Ether's realized average return was about 19% (a compound return of −7%; the gap is the volatility drag), just below the interval: unlike Bitcoin, Ether has delivered less than the equation requires over this window. Figure 3 shows Ether's volatility compressing alongside Bitcoin's; the equation says its required return has compressed with it.

Large altcoins. For the surviving assets in the largest size quartile outside the top two, volatility is about 80% and the all-cause death rate is 3.2% a year, mostly slow fades in which holders recover part of their value, so a mortality term of 1–3 points. The holder weight is not surveyed for individual coins; concentration among committed communities is plausibly higher than for Bitcoin, and we show an illustrative 0.10–0.30. The required return lands at about 39% a year (90% interval 27–60%; envelope 20–73%). The significant-asset cohorts of Section 5, which are the surviving large altcoins, delivered median compound

returns of -7% to -9% a year five to seven years after birth, an average return of roughly 25% once the volatility drag is added back: the typical large altcoin has fallen short of what its risk required by some 15 points a year on average, and lost money in compound terms.

Small coins. At the bottom of the ladder the mortality term stops being a footnote. The smallest quartile dies at 17.4% a year, with recovery close to nothing, so the death risk alone adds about 16 points to the required return before any compensation for volatility of 100% or more. The equation demands about 78% a year (90% interval 52–124%), and 39% at the very least. What was delivered is not observable in the price data, which the free tier does not carry for small coins; what is observable is that half of them are dead within 3.3 years of listing (Figure 8), and that even the surviving significant altcoins, a far stronger group, fell short. The pattern is the one documented for lottery-like stocks, where investors accept low average returns for a small chance of an extreme payoff (Barberis and Huang 2008); the graveyard prices that preference in this asset class, and for the half that die within three years the shortfall is the whole stake.

The altcoin holder weights and betas are illustrative and the realized comparisons cover only surviving assets with price history, which flatters them (Section 9); the direction of the result does not depend on either. As one moves down the size ladder the required return rises steeply, because volatility and death risk both rise, while realized returns fall short by more. Bitcoin is the one cryptoasset for which the two sides of the equation have met, which is a statement about Bitcoin's holder base and survival record, not about the equation.

8. FOR THE ALLOCATOR

The required return translates directly into practice. This section states the hurdle a diversified allocator should apply, inverts the equation into a sizing rule, places the result in a Black–Litterman frame, and then asks the question the numbers invite: whether, and for whom, Bitcoin is worth holding at all.

The hurdle. A diversified allocator should require of Bitcoin what its contribution to her portfolio's risk costs her: the risk-free rate plus beta times the equity premium, about **11–13% today**, plus some or all of the mortality term, 0.3–3 points, to the degree she judges crypto mortality systematic rather than diversifiable. The 2022 clustering of elite deaths (Terra, Celsius, FTX within seven months) argues it is partly systematic; we would charge 1–2 points. Call the hurdle 12–15%. This is not a forecast of what Bitcoin will do; it is the bar it must clear to belong in her portfolio at more than market weight. It is twice what it would have been before 2020, because Bitcoin now moves with equities.

The sizing rule. The required-return equation, inverted, sizes the position. If an investor's own view of Bitcoin's expected return is $\hat{\mu}$, her risk aversion γ, and Bitcoin's volatility σ, her weight should be

$$w^* \;=\; \frac{\hat{\mu} - r_f - \beta\,\mathrm{ERP} - \lambda(1-\delta)}{\gamma\sigma^2}. \tag{2}$$

Worked example: an investor with $\gamma = 3$ who believes our central forecast, $\hat{\mu} = 18\%$, at today's $\sigma = 42\%$ ($\sigma^2 \approx 0.18$) and a beta of 1.75: $w^* = (18 - 5 - 7 - 0.3)/(3 \times 0.18) \approx$ **11% of financial wealth**. That is aggressive by institutional standards, and a little over half of the 19% the average 2021 holder actually ran at far higher volatility. An investor who believes the age rule's 32% gets $w^* \approx 37\%$; the distance between those two positions is the practical cost of getting this paper's question wrong.

Buy and hold, or rebalance. The sizing rule, like equation (1), is written on the arithmetic return,

and who collects that return depends on what the investor does. A position bought and never rebalanced grows at Bitcoin's own compound rate, so its typical outcome is about 9% a year, whether it is the whole portfolio or a small slice of it; the 18% is the average across paths, lifted by a few very large ones. On that footing Bitcoin offers the long-term holder about what a global equity index does (about 8% compounded on this paper's inputs) at three times the volatility. A position rebalanced to a fixed weight is different: the slice contributes its arithmetic return, and the volatility drag is paid on the portfolio, not on Bitcoin. Five percent in Bitcoin, held against the S&P 500 and rebalanced, raises the portfolio's expected return by about 0.45 points a year and its volatility from 14.1% to 14.6%, which costs about 0.07 points in extra drag. The premium of this paper is therefore a rebalancer's premium; the buy-and-hold investor should plan on the compound figure. Table 6 sets the two rates side by side for Bitcoin and its natural comparisons. The drag is half the variance, so it grows with the square of volatility: it takes a point or so from equities and gold, half of Bitcoin's required return, and four-fifths of Ether's, whose compound figure ends up below Bitcoin's.

Table 6: Average and compound required returns, percent per year.

	Average	Volatility	Volatility drag	Compound	Sharpe ratio
Global equities (MSCI ACWI)	8.8	13.8	1.0	7.8	0.28
S&P 500	9.0	14.1	1.0	8.0	0.28
Gold	5.5	15	1.1	4.4	0.03
Bitcoin	18.0	42	8.8	9.2	0.31
Ether	26.0	64	20.5	5.5	0.33

Average: the arithmetic required return, in nominal US dollars; for the S&P 500 the 5% risk-free rate plus the 4% equity premium, for global equities the same premium scaled by their beta of 0.95 on the S&P 500, for gold Appendix E, for Bitcoin and Ether the central estimates of Table 5. These are required returns, not histories: over the sample the S&P 500 compounded at 15% a year and global equities at 11.5%. Volatility: for the two equity indices, measured on monthly total returns of the iShares MSCI ACWI ETF and the Vanguard S&P 500 ETF, January 2012 to April 2026, the sample of Section 5 (the all-cap Vanguard Total World Stock ETF gives 14.0%); the paper's estimates otherwise (the SPDR Gold Shares ETF gives 15.5% for gold over the same months). Drag: half the variance. Compound: average less drag, the rate at which a position that is bought and never rebalanced should be expected to grow. Sharpe ratio: average less the risk-free rate, divided by volatility.

The Black–Litterman translation. In a Black–Litterman portfolio, the tilt away from market weight is proportional to the gap between your expected return and the market-implied one (Black and Litterman 1992). This paper reduces that gap to observable disagreements: your volatility path, your view of holder-base integration (the path of w), and your mortality estimate against the market's option-implied one. An allocator with no disagreement on those three has no basis for holding Bitcoin away from its market weight, currently well under 1% of global financial wealth, however glorious its history.

Is it worth holding at all? For the smaller cryptoassets the answer is no: required returns of 39–78% against delivered returns that fall short or cannot be observed mean their holders have been underpaid on average, and the only coherent account of their demand is the lottery preference of Section 7. For Bitcoin the answer depends on who is asking. The 18% is what its concentrated holders require, and in equilibrium it is what they get, so for them it carries no alpha. A diversified investor is in a different position. She does not bear the concentration risk those holders are paid for, and at a market weight under 1% she barely bears the death risk either: a jump to zero would cost her less than 1% of wealth. Her hurdle is the 12–15% above, and the asset offers about 18%. The 3–6 point difference is a rent, the premium a still-concentrated holder base demands, available to anyone able to hold the asset as a small part of a diversified portfolio; that is Merton's

(1987) original observation about limited participation. The rent is what remains of the concentration term, and it shrinks as the holder base broadens.

Against equities, standing alone. Take the S&P 500. Its expected excess return is the equity premium, 4% (6% in the high case); its volatility was 14% a year over January 2012 to April 2026 (Table 6) and is 16–20% over the long history (Dimson, Marsh, and Staunton 2002). The ratio of excess return to volatility, the Sharpe ratio (Sharpe 1994), is therefore 0.20–0.29, and 0.30–0.43 in the high case. Bitcoin at an 18% required return, a 5% risk-free rate, and 42% volatility gives $(18-5)/42 \approx 0.31$, and about 0.27 once the mortality term, which is compensation for a loss and not a return, is taken out. If the required return is paid, Bitcoin offers about the same reward per unit of risk as equities on their recent volatility, and somewhat more on their long-run volatility; where there is an excess, it is the concentration premium. The choice of index changes little. A global index (MSCI ACWI) had a volatility of 13.8% over the same months against 14.1%, a correlation of 0.97 with the S&P 500, and a beta of 0.95 on it, so the same pricing gives it a premium of 3.8 points. Realized returns were much further apart: the S&P 500 compounded at about 15% a year over those months and the global index at about 11.5%. We do not carry that gap forward, for the reason we do not carry Bitcoin's history forward: a realized return is not a required one, and two indices with nearly the same risk cannot be required to pay very different premiums. Currency matters as well. These are US-dollar returns; measured in euros the global index was the less volatile of the two by more than a point (12.6% against 13.8%), because less of it is held in dollars. A reader who prefers the historical US premium of 7–8 points over bills (the familiar 10% nominal return of the last century) should raise both sides: equities' ratio rises to about 0.4–0.5, and Bitcoin's market term doubles with the premium, taking its required return to about 25% and its ratio to about 0.48. What has changed is the diversification argument. Bitcoin's correlation with equities, about 0.2 before 2020, has been about 0.5 since 2022 (Appendix A; Copestake, Furceri, and Terracciano 2026), so about two-fifths of its required return is now payment for a risk the equity investor already holds. What this paper removes is not the case for a small Bitcoin position but the case built on 32% and on zero correlation. At a central 18% on average, about 9% compounded, with a beta near 1.75, it is a high-beta risk asset with a shrinking concentration premium.

In Sharpe units. Dividing equation (1) by σ turns the required return into a required reward per unit of risk, and because $\beta = \rho\,\sigma/\sigma_m$ the market term simplifies:

$$\frac{\mu - r_f}{\sigma} \;=\; \rho\,\mathrm{SR}_m \;+\; \gamma\sigma\,w \;+\; \frac{\lambda(1-\delta)}{\sigma}, \tag{3}$$

where ρ is the asset's correlation with the market and SR_m the market's Sharpe ratio. For a fully diversified holder of an asset that cannot die the right side collapses to $\rho\,\mathrm{SR}_m$, the CAPM in Sharpe units: no asset can offer more reward per unit of risk than the market times its correlation with it. The concentration term, $\gamma\sigma w$, is what a holder adds when she owns a lot; σw is the position's contribution to her portfolio volatility, so Sharpe stops being scale-free. The mortality term shrinks with volatility, because a fixed expected loss is a smaller share of a larger risk budget. More generally, at any holder's optimum an asset's Sharpe ratio equals its correlation with her own portfolio times that portfolio's Sharpe ratio; concentration is simply what makes the correlation high. The numbers are telling. Today the market term (7 points over 42%, a correlation of about 0.5 at a market Sharpe ratio near 0.25–0.3) is about 0.15, the concentration term ($1.9 \times 0.42 \times 0.135$) about 0.11, and mortality (1.65 points over 42%) about 0.04: a required Sharpe ratio of about 0.30, the figure reached above by division. In 2013 the band midpoint implies about 0.8, and above 1 at its all-in

edge, a reward per unit of risk that no asset class delivers for long and Bitcoin did not. Once holders are diluted toward market weight ($w \to 0$) the requirement falls to $\rho\,\mathrm{SR}_m$ plus the mortality term, about 0.19 at today's volatility: an asset 0.5-correlated with equities should, in an integrated market, earn about half the market's reward per unit of risk plus a few points for death risk. Bitcoin's current Sharpe ratio exceeds that only because its holders are still concentrated, and integration should push it down toward equities' own. The same form answers the allocator's question directly: an asset improves a diversified portfolio's Sharpe ratio if and only if its own Sharpe ratio exceeds ρ times the portfolio's, here about 0.30 against about 0.13, so a small position adds today; the margin is the limited-participation rent plus the mortality allowance, and the rent vanishes as $w \to 0$ by construction. For gold, with $\rho \approx 0$ and a Sharpe ratio of a few hundredths, the improvement is negligible on expected-return grounds; whatever case gold has rests on tail behavior this condition does not price.

Appendix E applies the same condition to gold, where it reproduces a known result (a required return near the risk-free rate; Barro and Misra 2016), and spells out what "digital gold" would mean in return terms: about 14% a year at gold's volatility and holder dilution, because Bitcoin would still carry its beta and its death risk, and gold's 5–6% only if it shed both.

9. LIMITATIONS AND THE MEASUREMENT AGENDA

We list the paper's limits in the order a critical reader should weigh them.

1. **The holder weight before 2021 is bounded, not measured.** The portfolio-share surveys begin in 2021; everything earlier rests on market-clearing back-casts and option bounds, and the 2026 lower bound on an assumed weight for ETF and treasury investors (Appendix B). The single most valuable upgrade would be a measured w path: entity-adjusted on-chain concentration joined to holder-wealth estimates, the 2025 portfolio-share figures in the Candia et al. online appendix (not yet published in text at this writing), and the ETF share of total holdings.

2. **Coverage-onset selection in the graveyard.** Assets enter the panel only after becoming significant, so young-age hazards are understated. This biases the data *toward* Lindy, which makes the rejection conservative, and it leaves the elite estimates untouched; but population hazard *levels* at ages under two should be read as floors.

3. **Death timing and classification.** Feed and volume deaths lag economic death by months; four elite "deaths" are contested zombies (chains that operate far below their manic highs); one (ICP) is an artifact of a listing-day price spike. All classifications, dates, recovery fractions, and confidence flags are in Appendix C, and the headline elite hazard moves only from 3.5% to 3.15% when the contested cases are dropped.

4. **Bitcoin's own existential era is unmeasurable.** No cross-section existed in 2010–2014 to price the risk that Bitcoin itself was a dead end, so the early bands are wide by construction; the forward claim does not rest on them.

5. **Survey-to-wealth pairing.** The participation series is US-based and the wealth aggregates global; re-running the clearing calculations with US household wealth against US participation moves the

bands by amounts well inside their width (Appendix D). The surveys also measure crypto as a whole, which we read as Bitcoin's weight (Section 4); a back-cast on total crypto capitalization leaves the test unchanged and lowers the forecast by one point (Appendix D).

6. **Risk aversion is not measured for holders.** The upper end of the range for γ comes from studies of ordinary households; the lower end, 1, is our reading of surveys that find crypto owners more tolerant of risk (Hayashi and Routh 2025; Stix 2021), not a measurement.

7. **The holder weight is endogenous to beliefs.** A weight chosen on over-optimistic expectations satisfies the same first-order condition as one forced by segmentation, so the concentration term recovers the holder's subjective hurdle; whether that hurdle was a fair premium is adjudicated by realized returns (Section 5) and by the option-implied bound, not by the equation. The forecast assumes that the ETF-era weights are participation-driven; that assumption is what the 2031 test checks.

8. **The equation is complete only for the holder it describes.** The premia and yields it leaves out (Section 2) are not priced separately; the option-implied bound and the monthly test leave little room for a large net residual, but not none.

9. **The validation is on a survivor, and has limited power.** The monthly test does not reject the required return, and since 2018 the two agree within 10 points a year, but the standard error of that gap is 24 points. Early returns ran above the midpoint, Bitcoin is the winner of its cohort, the band is wide, the midpoint is a convention rather than an estimate, and the required return does not track the timing of returns (Section 5). The 2025 survey answer is below the model (Section 4).

10. **The market beta is unstable.** The rolling four-year beta has ranged from about zero to 3 and the correlation from zero to 0.55 (Appendix A); the forecast uses 1.5–2.0, and a return to pre-2020 correlations would lower the required return by several points.

11. **A short and noisy history.** Fourteen years of returns pin down an average only to within about 40 points a year (24 since 2018), however finely they are sliced. That is why the paper tests on monthly returns, leans on levels at measured dates, cross-sectional mortality, and two survey waves, and why its forecast is phrased so that a single new window can grade it.

12. **Illustrative inputs for the other assets.** The altcoin holder weights and betas of Section 7 and gold's holder weight in Appendix E are assumed ranges, not survey measurements, and gold's mortality allowance is a judgment; the required returns built on them are directional, and the Bitcoin estimate, which rests on measured inputs, does not depend on them.

10. CONCLUSION

The riddle was that Bitcoin seemed to carry three expected returns at once: 32% from its own tested history, 11–16% from the CAPM, 14–22% from its holders' mouths. The resolution begins with classification. Only the first was ever an expected return, a property of the price process. The CAPM range is a normative required return, the hurdle of a diversified investor. The third is what holders say they expect. What was missing is what those holders should require given the risks they actually bear, and this paper supplies it:

Merton's (1987) limited-participation pricing with measured inputs, a market term from a rolling beta, a concentration term from surveyed portfolios, and a mortality term from a census of 1,815 cryptoassets. The graveyard shows mortality was never the main story (hazard rises with age; the top-20 jump hazard is 0.3% a year). Concentration was: it supplied most of a required return that ran above 100% a year in Bitcoin's early life. And when that required return is tested month by month against what Bitcoin delivered, it is not rejected: early returns ran toward the top of its band, and since 2018 the two agree within about ten points a year. Bitcoin's history looks less like an anomaly than like a very large risk premium, paid.

What remains, if the asset is priced in equilibrium, is a required return built from observable things: 40-some percent volatility, a beta to equities near 1.75, a holder base diluting toward market weight, a small and datable mortality risk, and a risk-free rate. Those ingredients say about 18% a year on average for 2026–2031, with a 90% range of 15–22%, or about 9% compounded; the same equation, fed gold's inputs (Appendix E), returns the risk-free rate plus half a point. The age rule's 32% sits above the band, and the framework says why it should fail: it extrapolates a decline that was driven by integration, not by age, and integration has now raised one term while lowering two others. If Bitcoin instead repeats its age-rule 32%, then either it is still underpriced or a risk we measured as small is large, and the framework says which observables separate the two.

For practitioners. The message for a long-term portfolio has three parts. First, where Bitcoin is heading. As its holders diversify, equation (1) collapses to the CAPM, and the CAPM's advice is the familiar one (Tobin 1958; Sharpe 1964): hold the market portfolio and the risk-free asset, in proportions set by one's taste for risk. In that world Bitcoin is an ordinary high-beta asset that belongs in a portfolio at its market weight, under 1%, with a required return of 12–15%. Second, where it is today. Its holders are still concentrated, so it offers about 18% a year (15–22%), 3–6 points more than a diversified investor needs. An investor willing to hold more than market weight, and to rebalance, can collect that premium, up to about 11% of financial wealth at a risk aversion of 3; it is a rent that shrinks as ETFs broaden the holder base. Third, what a position that is bought and never rebalanced should expect: about 9% a year compounded, roughly what a global equity index offers, at three times the volatility, and not the 32% that Bitcoin's history extrapolates. Throughout, do not treat Bitcoin as uncorrelated (its beta to equities is about 1.75), and charge for mortality explicitly, 0.3–3 points, with option-implied crash intensity as the running check. A measured path for the holder weight is where the estimate can be sharpened most (Section 9); the forecast itself is graded in September 2031.

Beyond Bitcoin. The procedure is not specific to Bitcoin, and we offer it as a template for any new asset or asset class that is volatile, held by few, and able to fail. It has four steps: measure the asset's volatility and its beta to the investor's other wealth; find out who holds it and what share of their wealth they keep in it; count how often assets of its kind fail and what holders recover when they do; and add the three resulting terms to the risk-free rate. Section 7 applies it to Ether and smaller cryptoassets and Appendix E to gold. Those applications rest partly on assumed inputs, so the template is validated here on one asset only; tokenized assets, private credit and early-stage equity are natural next tests.

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Generative AI (Anthropic's Claude) was used under the author's direction to write and test the code in the replication package, to produce the figures from that code, and to draft and edit text. The research design, the curation of the death classifications, all interpretations and conclusions, and the final text are the author's, who has verified the outputs and takes full responsibility for the content. Generative AI was not used to generate or alter data.

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Data and code availability: the full replication package (panel-construction and survival scripts, the curated asset-level panel, all result tables, and every exhibit) is maintained at `https://github.com/apapadogiannis/bitcoin-required-return`. The pipeline ends with an audit script that recomputes and checks every quantitative claim in this paper. Underlying market data are the public Coin Metrics community dataset (snapshot ending May 24, 2026), which the package downloads from its source rather than redistributing.

A. MARKET RISK AND THE RISK-FREE RATE

The market term of equation (1) is Bitcoin's beta to the equity market times the equity premium, and it is added to the risk-free rate. The paper uses these three inputs in two places and treats them differently in each. Table 7 says what is used where; the rest of this appendix explains each choice.

Table 7: The risk-free rate, beta and the equity premium: what is used where.

	The 2026 estimate and forecast (Sections 4 and 6–8)	The path since 2012 (Section 5)
Risk-free rate	5.0%: the ten-year Treasury yield in 2026	the ten-year Treasury yield of each date
Beta	a range, 1.5–2.0	re-estimated every month on the previous four years
Equity premium	4% (6% as a high case)	4%, the same at every date

A.1 How beta is estimated. Beta is the slope of a regression of Bitcoin's monthly log return on that of the S&P 500, both measured month-end to month-end. The index is represented by the total return of the Vanguard S&P 500 ETF (VOO, adjusted close), whose month-end ratio to the index is constant to the third decimal over the sample. The S&P 500 is used as a proxy for the rest of the holder's wealth, not as an index Bitcoin belongs to: what equation (1) prices is Bitcoin's covariance with that wealth, and the true market portfolio of all assets is not observable (Roll 1977). A global equity index would serve as well; US large caps are most of it and the two move closely together. The equity premium of A.4 is the premium on this same index.

A.2 The beta used for 2026. The estimate and forecast use a range, 1.5–2.0, because the answer depends on how far back one looks and no window is clearly right. On the last four years beta is 1.75 (correlation 0.50); on the last five, the common convention for monthly data, 1.8; since 2016, 1.5; since 2012, 1.8. The range covers these. Shorter windows give less: 1.4 on three years and 1.3 on two. Each estimate has a standard error of 0.4 to 0.6, so all of them are within one standard error of one another, and the sampling error of any single estimate is larger than the range we use. Beta is an imprecise input, and the shorter windows suggest that the market term is, if anything, on the high side.

A.3 The beta used for the path. Section 5 needs a beta at every date since 2012, so there it is time-varying: at each month-end it is re-estimated on the previous four years (on all available data before four years exist). Figure 4 shows it with the corresponding correlation. Before 2020 the correlation was 0.0–0.3

and the beta was unstable, near 3 in 2013–2015 only because Bitcoin was so volatile, and about zero over the four years to early 2020. Since 2022 the correlation has been 0.3–0.6 and the beta 1.6–2.2. The rise in co-movement coincides with institutional entry and is documented independently by Copestake, Furceri, and Terracciano (2026).

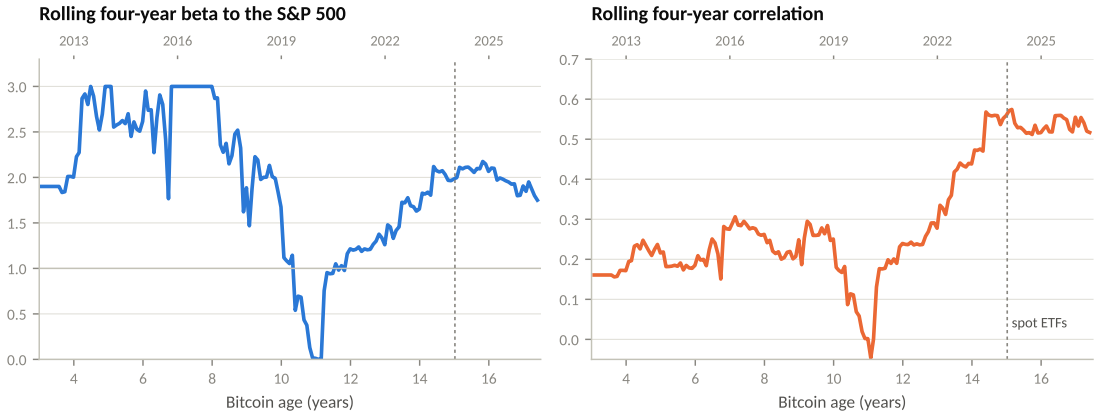


Figure 4: Bitcoin's rolling four-year beta and correlation with the S&P 500 (month-end log returns; expanding window before four years of data). The dotted line marks spot-ETF approval.

A.4 The equity premium. The equity premium is held at 4% at every date, with 6% as a high case. It is a forward-looking assumption, not an estimate that moves with the market: Fama and French (2002) put the expected US premium at 2.6–4.3% from fundamentals, the J.P. Morgan Asset Management (2026) assumptions imply about 3 points for US large caps over the ten-year yield, and realized premia have averaged more (Dimson, Marsh, and Staunton 2002). Section 8 reworks the comparison with equities at the historical 7–8 points. Each point of premium moves Bitcoin's required return by about 1.75 points.

A.5 The risk-free rate. The risk-free rate is the ten-year US Treasury yield, the usual choice for a multi-year horizon. For 2026 it is 5.0%. For the path of Section 5 it varies with the date: we use approximate mid-year yields every second year (1.8% in 2012, 2.5% in 2014, 1.8% in 2016, 2.9% in 2018, 0.9% in 2020, 3.0% in 2022, 4.3% in 2024 and 5.0% in 2026) and interpolate between those points. The survey comparison of Section 4 uses the yield at each survey date, 1.5% in 2021Q3 and 4.4% in 2025Q2.

A.6 Two approximations. First, for a holder with weight w in Bitcoin and the rest in the market, the exact market term is $(1-w)\beta$ times her price of market variance; at w of 8–19% the factor $(1-w)$ lowers the term by about a point, which the stated beta range absorbs. Second, a holder with risk aversion γ prices market variance at $\gamma\sigma_m^2$, which at $\gamma = 1$–2.8 and $\sigma_m = 16\%$ is 2.6–7.2%, a range that contains the 4% and 6% premia used; the market term and the concentration term are therefore priced consistently.

B. CONCENTRATION RISK

The holder weight w is the input the paper measures least directly, so this appendix shows how it is built (Figure 5). One number is measured: in 2021Q3 US households that owned any crypto held 19% of their financial assets in it on average (Candia, Coibion, Gorodnichenko, and Weber 2026), with the Austrian 2023 survey (OeNB 2025) as a cross-check. Everything else follows from market clearing, capitalization $= \bar{w}\times$ participating wealth, run backward and forward from that anchor.

B.1 Participation. The participant count is the share of US adults holding crypto in household surveys: 0.4% (2014) and 1.2% (2018) from the Federal Reserve's payment-choice surveys, 11% in 2021Q3, and 12% in mid-2023 and in 2025 from Candia et al. (2026), whose panel shows a dip of two to three points in between that we do not model. The 2021 level is the least settled. The Federal Reserve's SHED puts adults who used or held crypto at 12% in 2021; the Candia et al. panel reports 11% for 2022 but only about 3% earlier in 2021. We use 11% at the 2021Q3 anchor because a roughly flat path from there to 2025 is what the same survey's portfolio shares imply: they are similar in 2021 and 2025, which the back-cast reproduces only if participation is similar too. Other measures differ in level. Later SHED waves are lower (10% in 2022, 7% in 2023, 8% in 2024), and Pew's broader count of adults who have ever invested in, traded or used crypto is higher (16% in 2021, 17% in 2024). The path is interpolated log-linearly between survey dates; the Coin Metrics count of addresses holding a balance supplies the shape only before the first survey point. Panel (a) also shows the alternative of scaling the address count to the 2021 point, which overstates participation five-fold in 2018 (6.4% against 1.2%) because addresses multiply faster than people; it is not used.

B.2 Wealth behind the holders. Participating wealth is anchored at capitalization divided by 0.19 in 2021Q3 (about $4.3 trillion) and moved with the participant count times wealth per participant, a ramp from a quarter of the 2021 level in 2012 to 1.4 times by 2026 (Appendix D.5 replaces the ramp with US net worth per adult). From 2024 a second class of holders enters through the spot ETFs and public-company treasuries, about 2.5 million BTC by mid-2026 (roughly 1.25 million in US spot ETFs and 1.26 million in corporate treasuries, Strategy alone about 0.85 million), worth about $190 billion at the May 2026 price. Their investors are not "crypto owners" in household surveys and hold small weights; at 1.5% of wealth they represent about $13 trillion of participating wealth, at 3% about $6 trillion, against $7 trillion for survey-type holders (panel b).

B.3 The weight. Dividing capitalization by participating wealth gives the implied average weight (panel c): about 3% in 2013, 17% in 2017, 18% in 2021, and in 2026 either 23% for survey-type holders alone or 8% once the institutional channel is counted at 1.5%. The path spikes at the 2013, 2017 and 2021 peaks because capitalization moves with the price while participation is smooth; that is the feature Appendix D.4 documents as a negative predictor and the reason the paper treats the weight as an input to a date-by-date restriction rather than a signal. The band used in Section 5 runs from the diluted back-cast (lower edge) to the all-in holder of the option bound (0.9, upper edge); the 2026 range of Section 4, $w \in [0.08, 0.19]$, is the diluted value up to the surveyed value. Two omissions work in opposite directions and are left as they are: coins that are permanently lost (estimates of 3–4 million BTC) should be removed from capitalization, which would lower every weight by 15–20%, and the wealth of the roughly 15% of owners who hold more than half their portfolio in crypto is understated by an average, which would raise the weight of the price-setting holder if she is drawn from that group.

B.4 Risk aversion and the option bound. The other two ingredients of the concentration term are taken from outside the paper. Risk aversion is $\gamma \in [1, 2.8]$: the upper end from the household estimates of Section 2, the lower end from the evidence that crypto holders are more tolerant of risk (Section 4, which also explains why it is not calibrated to holders' stated expectations). The upper edge of the historical band is Martin's (2017) bound: for a holder with all her wealth in the asset and log utility ($\gamma w = 1$), the required excess return is the risk-neutral variance, 66–84 points a year on Deribit options over 2017–2022 (Almeida,

Grith, Miftachov, and Wang 2024) and 16–30 points at the implied volatilities of 40–55% observed since. It is a requirement for an all-in holder, not a forecast for the asset; the band's upper edge scales it down after 2020 as the holder base dilutes.

C. MORTALITY RISK

C.1 Hazard and age. A *hazard rate* answers a simple question: of the assets alive at a given age, what fraction die within the next year? If 100 three-year-old coins are alive in January and 12 are dead by December, the hazard at age three is 12% a year. Estimating hazards across ages, with assets that are still alive treated correctly as "censored" (we know they lived *at least* this long), is standard survival analysis; the machinery is the same used for patients, firms, and light bulbs.

The Lindy question. The age rule's most natural economic defense was mortality: perhaps Bitcoin's enormous returns were fair payment for a risk of sudden death that declines as the asset ages. This is the "Lindy effect," under which remaining life expectancy grows with age, so the hazard falls like k divided by age. If that were true with k near 5.6, the age rule would simply be survival compensation, rationally priced, with equation (1)'s mortality term doing all the work. The hypothesis is attractive, parameter-free in shape, and testable; testing it is the point of building the graveyard.

It fails. Table 8 shows the death rate by age bucket under the curated definition.

Table 8: Mortality hazard by age, curated death definition (1,815 assets, 520 deaths, 2010–2026).

Age (years)	Deaths	Exposure (asset-yrs)	Hazard (per yr)
0.1–0.5	6	693	0.9%
0.5–1	37	770	4.8%
1–2	111	1,213	9.2%
2–3	106	905	11.7%
3–4	82	661	12.4%
4–5	78	488	16.0%
5–7	63	502	12.6%
7–9	33	198	16.7%
9–12	4	53	7.5%

Buckets begin at age 0.1 (the first weeks of a listing are definitionally alive) and the sparse 12–17 bucket (7 asset-years, no deaths, Bitcoin the main occupant) is omitted; total exposure across all ages is about 5,640 asset-years.

The hazard *rises* with age through at least age nine. Formally, a Poisson regression of monthly deaths on log age, with calendar-year fixed effects to absorb market-wide waves such as crypto winters and standard errors clustered by asset, gives an age elasticity of **+0.42 (s.e. 0.04)**: each doubling of age *raises* the death rate by roughly a third, holding the calendar year fixed. Lindy requires -1. The estimate has the wrong sign by eleven standard errors, and it is robust: $+0.44$ to $+0.56$ across the three mechanical death definitions, $+0.65$ when the suspicious cluster of mid-2025 feed terminations is excluded, and positive within the elite subsample alone. Equivalently, the slope of log survival against log age, which equals k if the hazard really were k/age, gives **between about 0.2 and 0.5** depending on the death definition, **against the 5.6** the age rule would need. A 5.6 would have implied that a three-year-old cryptoasset had roughly a 1-in-20,000 chance of reaching Bitcoin's current age; the measured elite hazard, held constant, implies a 14-year

survival probability around 0.6. Cryptoassets age as young firms do, with a liability of adolescence (Brüderl and Schüssler 1990) followed by obsolescence, and not as the books and ideas for which the Lindy effect was named, whose life expectancy grows with age (Figure 6).

Where the data end. Three caveats belong next to that result. First, in the oldest populated bucket (ages 9–12) the raw hazard falls back to 7.5%, on 4 deaths and 53 asset-years; the rise is established through roughly age nine, the data's support ends near age twelve, and Bitcoin is 17.4. The Lindy rejection is therefore a statement about the asset class over its observed life span, not a measurement of Bitcoin's own current hazard, which is exactly why the pricing below uses the elite hazard *level* rather than an age extrapolation. Second, elite deaths cluster in calendar time (2022, 2024–25), with contagion episodes and delisting purges, not smoothly in age; the three jump-collapses in particular sit within six months of one another and were causally linked (Terra propagated through Celsius to FTX), so the Poisson confidence interval, which assumes independent events, understates the uncertainty. We treat 0.30% as the point estimate of a partly systematic process, which a diversified allocator may rationally charge for in part; Section 8 does so. Third, the 2024–25 delisting purge supplies 12 of the 36 elite deaths; treating every 2025 death as censoring lowers the all-cause elite hazard from 3.5% to 2.4% a year (CI 1.5–3.5%), so the mortality bound of $\leq$3 points used below is conservative under either reading.

C.2 Size and the elite. Table 1 in Section 3 sorts assets by peak trading volume; Figure 7 shows the same hazards with confidence intervals, Figure 8 the survival curves by size, and Figure 9 the timeline of elite deaths, with the 2022 cascade and the 2024–25 delisting purge. Excluding four contested cases, assets flagged dead by the drawdown rule but whose chains still operate (ICP, SUSHI, FLOW, NANO; Appendix C), the all-cause rate falls to 3.15% (CI 2.2–4.4%). The remaining elite deaths are slow fades: OMG, NEM, Bitcoin Gold and their like lost their listings over months or years, during which holders could exit at declining prices; losses of that kind live in an asset's volatility and average return, not in its jump hazard. The 520 dead assets are, in the taxonomy of the scarcity framework of Papadogiannis (2025), the "physically scarce but no longer desired" quadrant realized at scale: hard-coded supply caps outlived demand, and scarcity without desirability monetizes nothing.

C.3 The curated elite death table. All 36 deaths among the 219 ever-top-20 assets, with the classification each received. *Type*: jump = sudden, value-destroying collapse priced as a Poisson event; exchange = failure of the issuing exchange; fade = gradual abandonment and loss of listings; contested = flagged by a mechanical rule but the chain still operates. δ is the approximate fraction of pre-event value recoverable by a holder (last liquid price or successor claim as a share of pre-event value); it is an estimate, not an accounting figure. *Conf.* flags classification confidence (H/M/L). Dates are the curated economic death dates; mechanical rules alone would misdate LUNA, UST, and FTT, whose collapses are entered from the public record.

Asset	Death date	Type	δ	Note	Conf.
LTL	2014-12	fade	$\sim$0	minor 2013-era coin	M
FCT (Factom)	2021-07	fade	$\sim$0.05	sponsor company bankrupt 2020; chain stalled	H
LBC (LBRY)	2021-09	fade/regulatory	$\sim$0.05	SEC action 2021; project wound down	H
PPT (Populous)	2021-11	fade	$\sim$0	abandoned	M
ETP (Metaverse)	2022-01	fade	$\sim$0	abandoned	M

Asset	Death date	Type	δ	Note	Conf.
NCASH (Nucleus Vision)	2022-01	fade	$\sim$0	delisted majors 2021	H
ZB	2022-02	exchange	$\sim$0	ZB.com halted withdrawals 2022	H
WTC (Waltonchain)	2022-04	fade	$\sim$0	abandoned	H
UST (TerraUSD)	**2022-05-12**	**jump**	$\sim$0.02	algorithmic stablecoin depeg	H
LUNA (Terra)	**2022-05-12**	**jump**	$\sim$0.01	supply hyperinflation; LUNA2 airdrop gave partial recovery	H
QASH (Liquid)	2022-07	exchange	$\sim$0	exchange hacked 2021, absorbed into FTX	H
QC (Qcash)	2022-08	fade	$\sim$0	abandoned	M
FTT (FTX Token)	**2022-11-10**	**jump**	$\sim$0.05	FTX insolvency	H
MITH (Mithril)	2023-03	fade	$\sim$0	serial delistings	H
QSP (Quantstamp)	2023-05	fade	$\sim$0.02	token abandoned	M
SEELE	2023-07	fade	$\sim$0	abandoned	M
VTC (Vertcoin)	2023-08	fade	$\sim$0.02	lost major listings	M
MDA (Moeda)	2023-09	fade	$\sim$0	abandoned	M
BTS (BitShares)	2024-06	fade	$\sim$0.02	community remnant only	M
AE (Aeternity)	2024-10	fade	$\sim$0.02	team wind-down	M
BCN (Bytecoin)	2024-10	fade	$\sim$0	long dormant; delisted majors years earlier	H
OMG (OMG Network)	2025-02	fade	$\sim$0.02	Binance delisting; development ceased	H
KEY (SelfKey)	2025-04	fade	$\sim$0	abandoned	M
LOOM (Loom Network)	2025-05	fade	$\sim$0	abandoned	H
XEM (NEM)	2025-05	fade	$\sim$0.05	major delistings; earlier opt-in XYM airdrop (2021)	H
BTG (Bitcoin Gold)	2025-06	fade	$\sim$0.02	Binance delisting	H
LINA (Linear)	2025-06	fade	$\sim$0	project shut down	M
PDA (PlayDapp)	2025-06	fade	$\sim$0.02	post-2024-exploit decline (PLA$\rightarrow$PDA swap preceded)	M
YFII (DFI.Money)	2025-07	fade	$\sim$0	abandoned fork	M
FLOW	2025-09	**contested**	n/a	chain operating; $>$99% below ATH	H
MXC	2025-09	fade	$\sim$0.02	rebranding/decline	L
STMX (StormX)	2025-10	fade	$\sim$0	abandoned	M
NANO	2025-12	**contested**	n/a	active community; lost major listings	M
ICP (Internet Computer)	2026-01	**contested**	n/a	alive, top-100; "death" is an artifact of the listing-day price-spike ATH	H
HIFI	2026-01	fade	$\sim$0	delisted	L
SUSHI (SushiSwap)	2026-03	**contested**	n/a	protocol operating; $>$99% below ATH	H

Headline sensitivities. All 36 deaths: hazard 3.54%/yr (CI 2.48–4.90). Excluding the four contested cases: 3.15%/yr (CI 2.15–4.44). Jump-collapses only (LUNA, UST, FTT): 0.30%/yr (CI 0.06–0.86). In the ever-top-50 universe the jump set adds SRM (Serum) and ANC (Anchor), both 2022 casualties of the same two cascades, for 0.23%/yr.

Migrations excluded as deaths (holders received successor value): MATIC$\rightarrow$POL, EOS$\rightarrow$Vaulta (A), FTM$\rightarrow$S (Sonic), GNT$\rightarrow$GLM, RNDR$\rightarrow$RENDER, BTT redenomination, NU$\rightarrow$T (Threshold merger), STRAT$\rightarrow$STRAX, MAID$\rightarrow$eMAID, HT$\rightarrow$HTX, REP$\rightarrow$REPv2, AGIX$\rightarrow$FET (ASI merger), XZC$\rightarrow$FIRO, TOMO$\rightarrow$VIC, NPXS$\rightarrow$PUNDIX, BNX redenomination, and DGD, whose dissolution refunded the ETH treasury to holders: a death with δ near 1, the mirror image of LUNA.

C.4 The Lindy regression. Poisson regression of monthly death indicators on log age, offset $\log(1/12)$, standard errors clustered by asset. Lindy (hazard $= k/\text{age}$) requires a log-age coefficient of -1.

Death definition	Deaths	γ_{age}, no FE	γ_{age}, year FE
Strict (feed termination)	234	$+0.56$** (0.07)	$+0.44$** (0.06)
Broad (feed + price)	263	$+0.69$** (0.07)	$+0.56$** (0.07)
Abandonment (all mechanical)	588	–	$+0.51$** (0.04)
Curated (final)	**520**	–	$+0.42$** **(0.04)**
Curated, 2025 feed-wall removed	–	–	$+0.65$** (0.08)
Curated, birth-cohort FE in place of year FE	520	–	$+1.62$** (0.10)
Elite (ever top-20) subsample	36	–	$+0.68$** (0.21)

Standard errors in parentheses, clustered by asset. ** significant at the 1% level; * at the 5% level. Every coefficient differs from the Lindy value of -1 at the 1% level.

The log-survival–log-age slope over ages 1–10 runs from -0.21 (strict) to -0.49 (curated), i.e. a cumulative Pareto constant $\hat{k} \approx 0.2$–0.5 across definitions, an order of magnitude below the 5.6 the age rule would require. Two further robustness checks address the standard identification worry that an age effect might be a disguised cohort effect (age, cohort, and period are collinear, so only two can enter at once): replacing calendar-period fixed effects with birth-cohort fixed effects *raises* the age elasticity to $+1.62$ (s.e. 0.10), so the rejection of Lindy is not an artifact of the period controls, and the period-FE $+0.42$ is the conservative figure. Parametrically, a Weibull fit to the curated durations gives shape $\rho = 1.57$ (95% CI 1.47–1.67), where $\rho > 1$ means a hazard rising with age and Lindy requires $\rho < 1$; a log-logistic fit (slightly preferred by AIC, 3,342 vs 3,371) implies a hump-shaped hazard that rises through the first decade of life. The implied median lifetime of a core-universe cryptoasset is 6.4 years under both the Weibull and the Kaplan–Meier estimates.

C.5 Death-definition sensitivities. Volume-collapse deaths require an absolute floor ($250k/day trailing volume) in addition to the 1%-of-peak rule; without the floor, 2021-mania base effects falsely kill operating mid-caps. The elite hazard is 3.54% with all curated deaths, 3.15% excluding the four contested zombies, and 2.36% (CI 1.51–3.51%) treating all twelve 2025 deaths as censoring; the Lindy rejection holds under every variant (C.2).

C.6 Forward-looking hazards the graveyard cannot see: the quantum threat. The hazard measured here is a backward-looking frequency, and some risks have never yet fired for anyone. The clearest is quantum computing: a cryptographically relevant quantum computer running Shor's algorithm would break the elliptic-curve signatures that secure Bitcoin, and the event would be exactly what the required-return equation calls a jump, a λ-event with its own loss fraction. The model therefore *covers* the quantum threat by construction; what the graveyard cannot do is price it, because the realized death count contains no quantum events. Two observations discipline it. First, the loss given the event is scenario-dependent, not automatically total: roughly a quarter of Bitcoin's supply sits in addresses with exposed public keys, including the presumed-Satoshi coins (Deloitte's standing estimate is about 4 million BTC), and is the immediately vulnerable part, while a telegraphed advance in quantum hardware would allow a migration to post-quantum signatures that preserves most other value; so δ ranges from near zero (a surprise break) to substantial (an orderly transition). Second, if the market prices a material quantum hazard, it must appear where all jump risk appears, in deep out-of-the-money put prices. The option-implied crash intensity is therefore the running referee between the graveyard's 0.3% and any larger forward-looking hazard, quantum included; at this writing it shows no such gap. An investor who privately assigns, say, a 1% annual probability to a disor-

derly quantum break with half of value lost should simply add 0.5 points to the mortality term; the equation accepts the view without modification.

D. ASSEMBLY AND VALIDATION

D.1 Propagating input uncertainty. The 13–25% envelope of Section 4 sums the extremes of four input ranges: beta 1.5–2.0 on a 4% equity premium; risk aversion $\gamma \in [1, 2.8]$; holder weight $w \in [0.08, 0.19]$; mortality term 0.3–3 points; with the risk-free rate at 5.0% and $\sigma = 0.42$. To obtain a distribution rather than an envelope we draw each input independently and uniformly over its range, 400,000 draws, and compute the required return $r_f + \beta\,\mathrm{ERP} + \gamma\sigma^2 w + \lambda(1-\delta)$ for each. The median is 18.0%, the central 68% interval 16.2–20.2%, and the 90% interval 15.3–21.6%. The uniform assumption is deliberately uninformative; any distribution concentrated toward the interior of the ranges tightens the result. Narrowing the mortality term to 0.3–1.5 points moves the median to 17.2% and the 90% interval to 14.8–20.7%; narrowing the holder weight to 0.12–0.19 gives 18.8% and 15.9–22.0%; the household range for risk aversion, 1.7–2.8, gives 18.9% and 16.2–22.1%. Volatility is held at 42% in the baseline; drawing it uniformly from 35–50% leaves the median at 18.1% (15.1–22.4%), and from the option-implied 40–55% raises it to 19.2% (15.8–24.4%). The same procedure gives Table 5 and the gold figures of Appendix E (beta uniform on $[-0.1, 0.1]$, σ on $[0.13, 0.20]$, w on $[0.02, 0.10]$, mortality on 0–0.3 points: median 5.5%, 90% interval 5.0–6.0%) and Bitcoin's digital-gold endpoint ($\sigma = 0.15$, w on $[0.02, 0.10]$, beta and mortality at Bitcoin's ranges: median 13.9%, 90% interval 12.3–15.5%). The script is `23_uncertainty.py` in the replication package.

D.2 Units. Equation (1) is a condition on the arithmetic average return μ, the drift. The compound growth rate of a lognormal price is $\mu - \sigma^2/2$, so the same asset has an average return and a CAGR that differ by half its variance: 9 points at 42% volatility, 32 at 80%, over 100 at 150%. Section 5 therefore compares the required return with the realized average return, estimated in two ways: in the monthly test as twelve times the mean simple monthly return, which estimates it directly, and over five-year windows as twelve times the mean monthly log return plus half the annualized variance of monthly log returns. Comparing with the CAGR instead would understate delivery by the drag; over the 2021Q3–May 2026 window the CAGR is 13% and the average 27%.

D.3 The test, monthly and in five-year windows. *The band.* At each month-end the lower edge is $r_f + 0.85\,\beta\,\mathrm{ERP} + \sigma^2 w_{lo} + 0.3$ points, with $\gamma = 1$ and w_{lo} the back-cast weight including the institutional channel. The upper edge is the larger of two numbers. One is the holder equation at its high inputs, $r_f + 1.15\,\beta\,\mathrm{ERP} + 2.8\,\sigma^2 w_{hi} + 3$ points, with w_{hi} the back-cast weight without the institutional channel. The other is the option bound, $r_f + \kappa\sigma^2$, the requirement of a holder with $\gamma w = \kappa$: κ is 1 (all-in, log utility) through 2020 and is lowered to 0.6 in 2022, 0.15 in 2025 and 0.10 in 2026, interpolated in between. That schedule is a judgment about how quickly the most concentrated price-setting trader was diluted, not a measurement. The option bound sets the upper edge in 68% of the months, for the last time in February 2023; from 2024 the upper edge is the holder equation. The 15% either way on the market term stands for the sampling error of beta. The midpoint is the average of the two edges. *Monthly.* At each month-end t from January 2012 to April 2026 the required return μ_t^{req} is converted to an expected one-month simple return, $e^{\mu_t^{req}/12} - 1$, and subtracted from Bitcoin's simple return over the following month; Table 4 reports twelve times the average difference and its standard error (172 observations; heteroskedasticity- and autocorrelation-robust errors with six lags give the same t-statistics to within 0.3). The age rule enters in

its one-month form, $((A+1/12)/A)^{5.62}-1$ at age A. Returns are heavily right-skewed: the largest month (November 2013, +451%) contributes 31 points to the full-sample mean of 118, and excluding calendar 2013 the delivered average is 69%, the midpoint 61%, and the gaps +48 (lower edge, $t=2.4$), +8 (midpoint) and −35 (upper edge). *Five-year windows.* Table 10 lists the band and the outcome at two-year intervals. Over the 113 overlapping windows starting January 2012 to May 2021, the realized average return minus the required return at the window's start averages +70 points at the band's lower edge, −49 at its upper edge, and +11 at its midpoint (median +23), and lies inside the band in 56% of windows. As a forecast of each window the midpoint has a mean absolute error of 48 points and a correlation with the realized average of 0.21; the age rule has 21 points and 0.83, and is closer in 88% of windows. The window comparison is kinder to the midpoint than the monthly test over the same months (a gap of +11 against +75) because it holds the requirement at its starting value for five years while the requirement was falling, and because overlapping windows weight the middle of the sample; we treat it as a description of the path and the monthly comparison as the test. Both say the same thing about timing, for the reason in D.4: a rate can be the correct required return and a poor forecast of when returns arrive.

Table 10: Required return versus realized average return, by starting date.

Start	Volatility	Beta	Required band	Midpoint	Realized average	Realized CAGR	Age rule
2013	151%	3.0	20–231%	126%	144%	135%	132%
2015	115%	2.9	17–135%	76%	99%	103%	90%
2017	80%	2.2	21–67%	44%	76%	51%	68%
2019	90%	0.7	24–82%	53%	58%	42%	55%
2021Q3	77%	1.3	16–47%	31%	27%	13%	45%
2023	66%	2.1	15–29%	22%	–	–	40%
2026	42%	1.75	13–27%	20%	–	–	33%

Rows are mid-year except 2021Q3, the survey date, whose realized figures run to May 2026 (4.7 years); other realized figures cover the following five years. The required band runs from the survey-anchored back-cast of the holder weight (lower edge) to the all-in option bound (upper edge). Volatility and beta are trailing estimates at the start date. For 2026 the propagated estimate of Section 4 is 18% (90% interval 15–22%). Age rule: five-year form.

D.4 The concentration quantity is not a timing signal. Forward h-year log growth is regressed on the real-time concentration quantity $x=\sigma^2 w$ (with w the back-cast weight) and the age-rule regressor $z=\ln(1+h/A)/h$; heteroskedasticity- and autocorrelation-robust standard errors with $12h+6$ lags; overlapping monthly windows from 2012 (independent observations ≈ 2–4, so the t-statistics are descriptive).

Horizon	x alone	z alone	Joint: x	Joint: z	R^2 (joint)
3y	−0.8 (t −4.7)	+4.9 (t 5.0)	−0.8	+5.1	0.57
5y	−0.8 (t −5.9)	+7.0 (t 13.5)	−0.8	+6.8	0.82

The concentration quantity enters with a negative sign: it is highest at market peaks, because capitalization sits in the numerator of w and trailing variance is highest after a run-up, and returns after peaks are low. The age regressor reproduces the power-law exponent (5–7). This is why the paper treats the required-return equation as a date-by-date restriction and never as a forecasting variable; it is also why the age rule's in-sample fit should not be read as evidence of a priced mechanism.

D.5 Wealth-path robustness. The back-cast's wealth-per-participant ramp (a quarter of the 2021 level in 2012, 1.4 times by 2026) is a judgment about who held Bitcoin. Replacing it with the growth of US

household net worth per adult (Federal Reserve Z.1, about $80T in 2013 to $170T in 2026), so that early holders are assumed as wealthy as the average American, lowers the early back-cast weight and the band's lower edge: 2013 becomes 18–231% (baseline 20–231%), 2017 (21–67%) and 2021Q3 (16–47%, the anchor) are unchanged, and 2026 becomes 13–30% (baseline 13–27%). The average gap between the realized five-year average and the band midpoint moves from $+11$ to $+12$ points, and the monthly gap is unchanged at $+51$.

D.6 A crypto-wide back-cast. The surveys measure the share of crypto as a whole in owners' financial assets, and Section 4 reads that share as Bitcoin's weight because the other large cryptoassets move about one for one with Bitcoin. The back-cast of Appendix B, however, clears Bitcoin's own capitalization against participating wealth, so it assumes Bitcoin's share of the asset class stayed at its 2021Q3 value. It did not: in our panel (assets with a reported capitalization, excluding stablecoins and wrapped or tokenized claims) that share was 95% in 2013, 38% in mid-2017, 48% in 2021Q3 and 74% in May 2026. Clearing the whole asset class instead, with the same anchor, participation path and wealth ramp, gives a holder weight of 1.7% in 2013 (baseline 3.4%), 21% in 2017 (17%), 19% in 2021Q3 (the anchor), 9% in 2023 (11%) and, in 2026, 15% without the institutional channel (baseline 23%) and 8% with it (8%). The monthly test is unchanged: the gap between delivered returns and the midpoint is $+52$ points over the full sample (baseline $+51$), $+111$ over 2012–17 ($+108$) and $+9$ over 2018–26 ($+10$). The forecast is where it matters. The 2026 band becomes 13–23% (baseline 13–27%), and if the upper bound on w in Section 4 is set at this 15% instead of 19%, the central estimate falls from 18.0% to 17.4% (90% interval 15.0–20.3%). We keep 19% as the upper bound in the headline for two reasons. It is a survey number, and the 2025 wave of the same survey reports portfolio shares similar to 2021's, which is what the baseline back-cast implies and this one does not. And the panel behind this back-cast omits some large assets, so it overstates Bitcoin's share of the class today. We read the result as a downside sensitivity of about half a point.

E. EXTENSIONS

E.1 Gold, as a check on the equation. Gold is the natural comparison, both because Bitcoin is so often called digital gold and because gold is the one other large asset that pays no income, so the whole of its return must come from price. Equation (1) prices it with the same four inputs, every one of which sits at the opposite end of the range from Bitcoin's. Market risk: gold's correlation with equities is close to zero over long samples (Erb and Harvey 2013), so the beta term is a few tenths of a point either way ($\beta \in [-0.1, 0.1]$). Concentration: gold is the most widely held asset on earth, about 224,000 tonnes above ground, of which 44% is jewelry, 18% central-bank reserves, and 23% private bars, coins, and ETFs (World Gold Council 2026), worth about $29 trillion at today's price, some 6% of global household wealth; its typical holder is a diversified investor with a few percent in it, not an enthusiast with a fifth. With volatility of 13–20% (13.1% over 1836–2011 in Barro and Misra 2016; higher in the last two years), a holder weight of 2–10%, and $\gamma \in [1, 2.8]$, the concentration term is 0–1.1 points, median 0.3. Mortality: gold has no failure mode that destroys the asset; the closest analogues are sovereign restrictions, such as the United States' compulsory purchase of private gold in 1933 at the old official price before the dollar was devalued, which are jurisdiction risks to a holder rather than deaths of the asset, and we allow 0–0.3 points. The equation returns a required return for gold of about 5.5% a year (90% interval 5.0–6.0%; envelope 4.6–6.8%; Appendix D.1): the risk-free rate plus half a point, a Sharpe ratio near zero. This is a reproduced result, and that is its value here:

Barro and Misra (2016) find that gold's real price appreciation averaged 1.1% a year over 175 years, with a covariance with consumption indistinguishable from zero, and conclude that its expected return should sit close to the risk-free rate; the equation reaches the same place from observable portfolios rather than from a consumption model. Gold's recent history is instructive. From the end of 2023 to November 2026 gold compounded at about 27% a year (from about $2,060 to about $4,040 an ounce, with a gain of some 65% in 2025 alone and a flat twelve months since; LBMA 2026), roughly twenty points a year above anything its holders could have required. That excess is news, not a payment for risk: central-bank purchases, fiscal and geopolitical developments, and ETF inflows the market had not priced. It is also a reminder that three years of realized returns say little about a required return, for gold as for Bitcoin. In equilibrium the equation puts gold's expected return from here back at 5–6% nominal; extrapolating 2024–25 would be the gold version of the age rule.

E.2 What "digital gold" would mean. The comparison also fixes the phrase in return terms. If Bitcoin's volatility fell to gold's 15% and its holder base diluted to gold's few percent per holder, the concentration term would all but vanish, as gold's has, and what remained would be the two terms gold does not carry: a beta of 1.5–2.0 and a mortality term of 0.3–3 points. The equation then gives about 14% a year (90% interval 12–16%; Appendix D.1). Only if those two also converged to gold's, with Bitcoin as uncorrelated with equities as gold and as immune to failure, would its required return reach gold's 5–6%. On the evidence of Appendix A, Bitcoin has been moving away from gold in that respect, not toward it: integration has lowered its concentration premium and raised its market risk. Digital gold names a required return near the risk-free rate; Bitcoin's is converging instead on that of a high-beta equity.

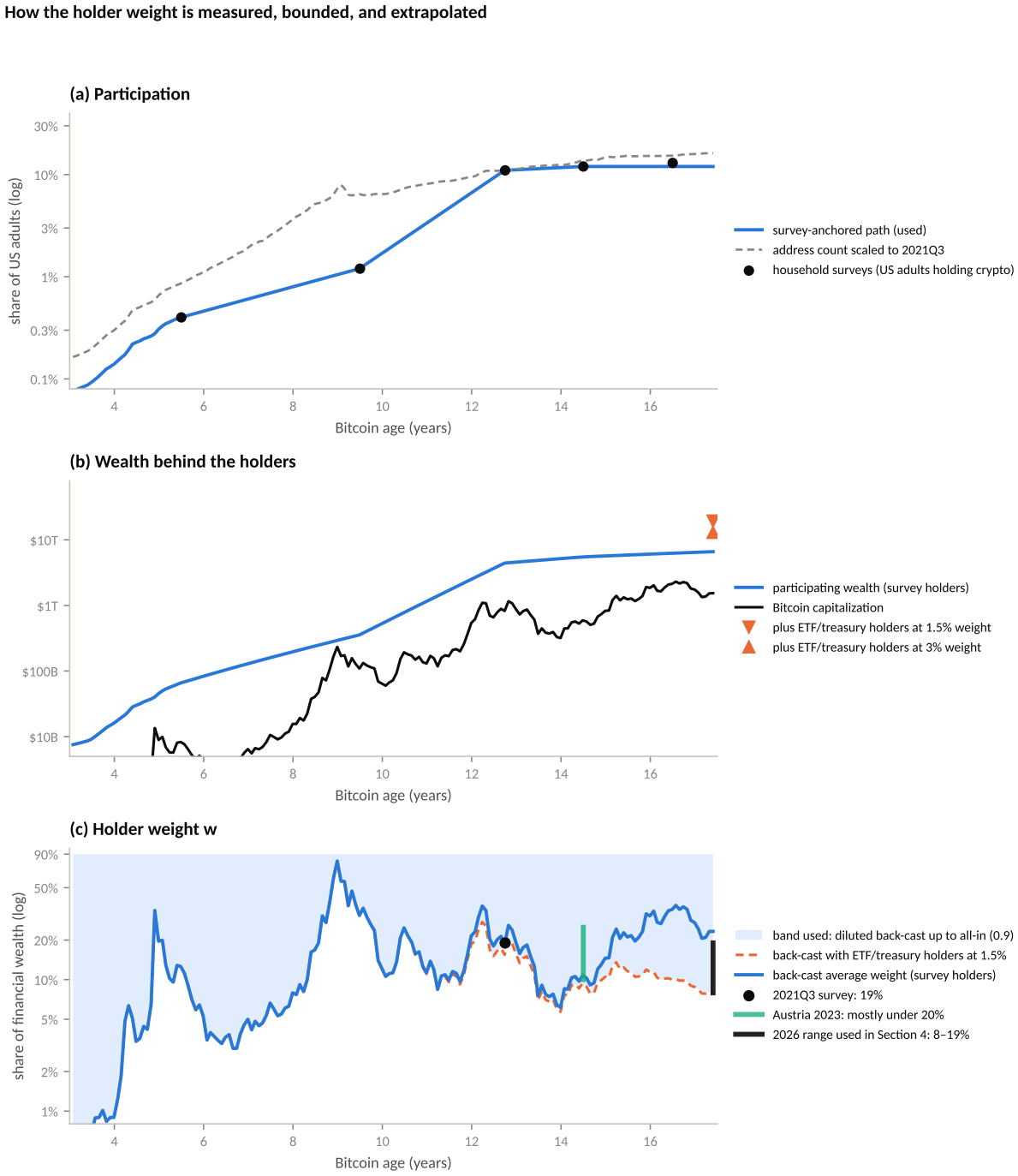


Figure 5: Constructing the holder weight. (a) Share of US adults holding crypto: survey points, the log-linear path used, and the address-count alternative not used. (b) Participating wealth of survey-type holders, Bitcoin's capitalization, and the 2026 additions for ETF and treasury investors at 1.5% and 3% weights. (c) The implied average weight, the 2021Q3 survey value, the Austrian cross-check, the diluted back-cast, and the band used in Section 5; all panels on logarithmic scales.

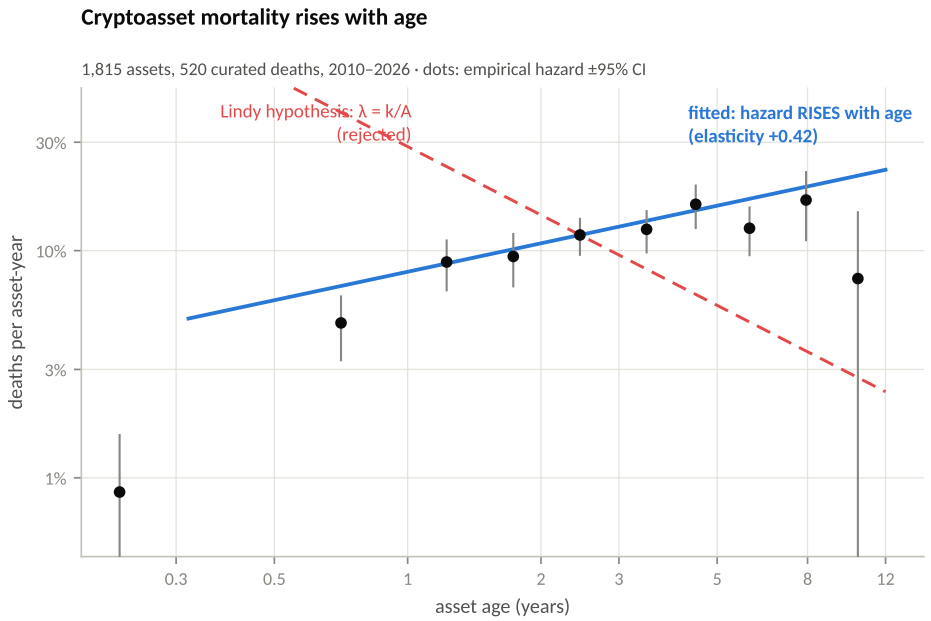


Figure 6: Mortality hazard versus age (log–log). The fitted line uses the period-fixed-effects elasticity of $+0.42$; the dashed line is the Lindy shape $\lambda = k/A$.

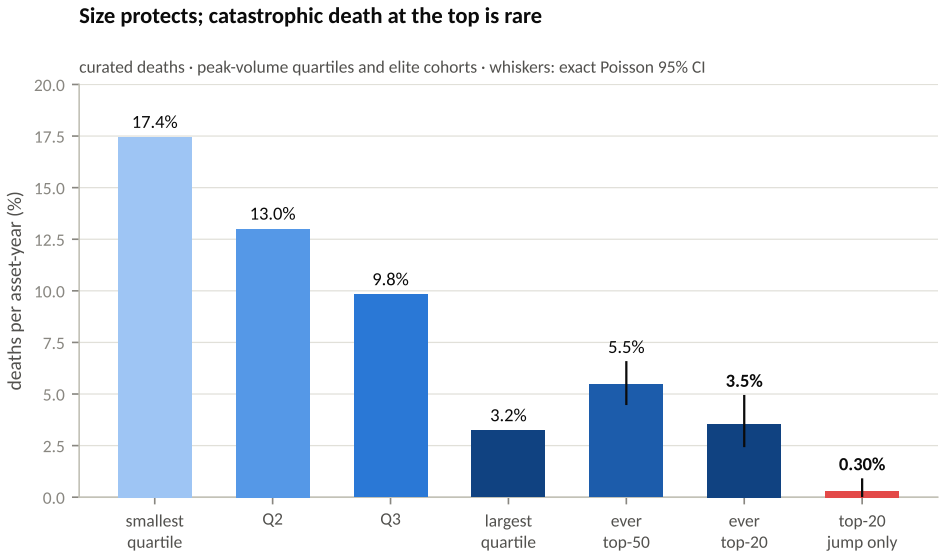


Figure 7: Hazard by peak-volume quartile and for the elite cohorts; whiskers are exact Poisson 95% intervals.

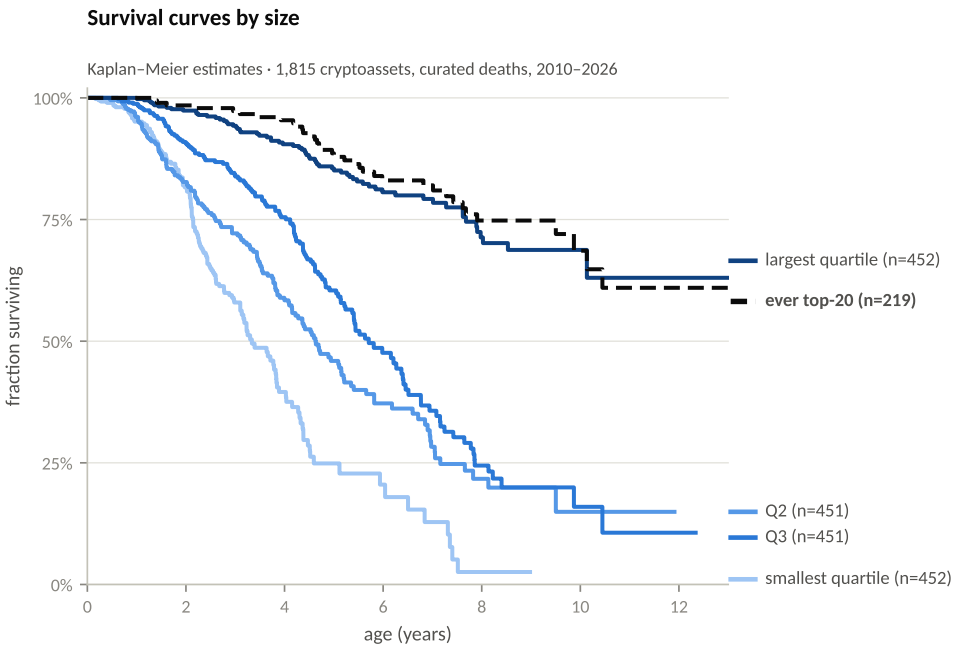


Figure 8: Kaplan–Meier survival curves by peak-volume quartile, with the ever-top-20 cohort (dashed). Half of the smallest quartile is dead by age 3.3; the largest quartile and the elite track each other and remain above 60% survival past age 10.

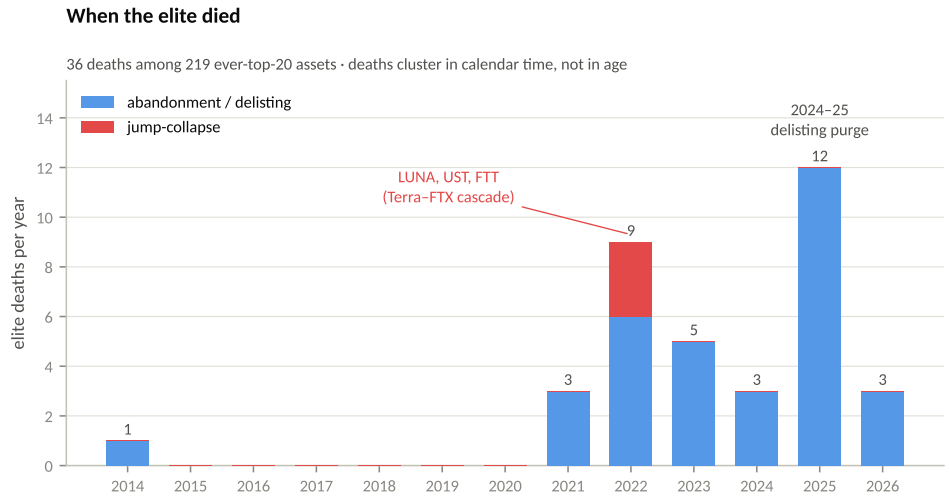


Figure 9: When the elite died: 36 deaths among 219 ever-top-20 assets, by year and type.